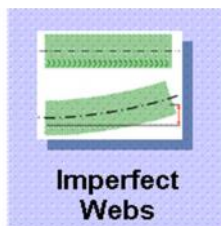
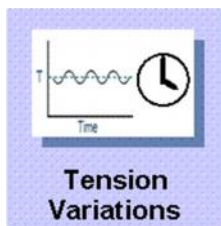




Solutions to Web Handling Problems



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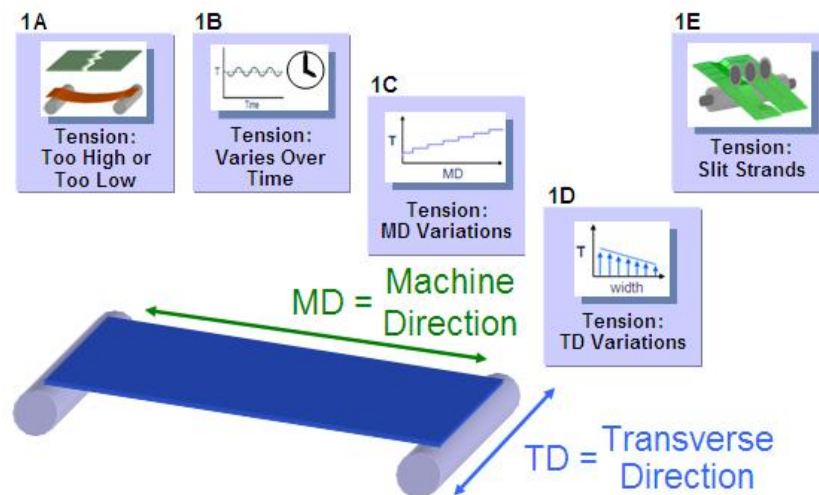
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I. Solutions to Web Tensioning Problems



Initial Steps in Solving Tension Problems

With this information in hand, you are in a good position to assess the causes and solutions to your tension problems.

1. Know your product mechanical properties
 - Cross-sectional area: thickness, width
 - Elasticity: Young's modulus of elasticity (or spring constant when thickness is difficult to define or measure), yield or break point.
 - Traction: coefficient of friction (COF) to roller surfaces, COF of your product front to back sides, porosity, surface roughness, adhesion and peel force to roller surfaces, peel force of your product at unwinding front to back sides.
 - Bagginess and Skew: understand the crossweb length variations of product and the tensioning force required to pull out the web flat.
 - Splicing: know the square-ness of your splicing methods.

2. Know your equipment geometry
 - Rollers: diameters, diameter variations, runout, balance
 - Roller alignment: level, tram (squareness to machine centerline)
 - Diameter uniformity vs. width
 - Nipping systems: force of nip load, geometry of nipping rollers, footprint of nipping rollers.

3. Map out your process tension
 - Fill in the following table for all tension zones

Start of Tension Zone	End of Tension Zone	Control Method for Zone
What driven, clutched, or braked element <u>starts</u> the tension zone?	What driven, clutched, or braked element <u>ends</u> the tension zone?	How is tension created and controlled in the tension zone?

- Process elevation diagram showing all rollers.
- Note all the tensioning elements (wherever a roller or roll is driven, clutched, or braked).

- Every set of two drive points creates a tension zone (no matter how short).
- Determine the pacer section of the line. (If needed, ask the controls support group for this info.)
- Note all tension feedback elements (usually load cell or dancer rollers, but also a hanging loops, vacuum boxes, tension ports in air turns, or accumulators).
- Zone without feedback must be either torque control or draw control.
- In torque control zones, look for or ask about use of diameter measurement feedback.

4. Collect process data

- Start with one product.
- Find the tension, torque, or speed set points that control each tension zone.
(This may be as easy as looking at the control panel. It may be quite difficult to chase down all clutch and brake settings, air pressure to dancers, and speed ratio of draw control zones.)
- Note the target process speed and acceleration / deceleration times or rates.
- Note any inherent web path length changes in the process, such as turret winders, accumulators, and wrapped nip rollers that open and close.
- Note if the web path changes by product or process, especially if alternate web paths bypass tensioning or tension feedback elements.
- Repeat this section for other products.

1A: Tension is Too High or Too Low / Slack

Definition	The total web tension is above the desired level (too high) or <i>below the desired level (too low) or the web is physically slack (no tension).</i> <i>(Comments in this section related to low and slack tension are in italics.)</i>
How to Measure	➤ Web tension (high or low) is measured with a load cell roller. High: ➤ Web breaks. ➤ Yielding may be difficult to measure except by excessive necking (width loss) and unreasonable speed increases from elongation. Low: ➤ Look for large sag in horizontal spans and non-turning idler rollers.
Problems with <u>High</u> Tension	✓ Web necks to unacceptable reduced width, leading to: > loss of edge trim > mismatched widths at laminating > loss of edge margin at coating > roll sidewall misalignment at winding (Necking is more significant for low modulus webs and webs with high Poisson's ratios (some, but not all tissues, textiles, nonwovens, and film). Necking is especially troublesome for webs of thermoplastic materials at elevated temperatures.) ✓ Web yields. ✓ Web or splice breaks. ✓ Print or die cut patterns are out of tolerance, both shape and phase. ✓ Wrinkle sensitivity increases with added web-roller friction. ✓ Winding or unwinding rolls cinching and telescope. ✓ Wound rolls are too tight, leading to tight roll defects, such as blocking, core crushing, starring, spoking, and yielding (that creates bagginess). ✓ Crashing on air floatation systems (air turns, floatation nozzles).
Problems with <u>Low</u> or Slack Tension	✓ <i>Web sags excessively in horizontal spans.</i> ✓ <i>Web fails to peel and wraps rollers, breaking the web or damaging the equipment.</i> ✓ <i>Web slips on idlers, causing scratches, lateral shifting, and guiding system failure.</i> ✓ <i>Web slips on driven rollers, especially unnipped drive rollers.</i> ✓ <i>Web slips on pacer drive rollers causes a loss of speed control.</i> ✓ <i>Web slips on follower drive rollers causes a loss of tension control or draw ratio.</i> ✓ <i>Web sails laterally in air floatation ovens and turns.</i> ✓ <i>Web flutters, especially in long spans.</i> ✓ <i>Wound rolls are too loose, causing telescoping in handling, winding, or unwinding.</i>

Mechanisms and Root Causes	High and low tensions are often flip sides of the same root cause. There are few that are specific to high or low and don't really have a reverse mode (such as there is no opposite of nipping). Root causes associated with BOTH high and low / slack tension. ▪ Torque applied to unwinding, winding rolls or intermediate rollers are too high or too low. ▪ Draw or speed ratios applied between driven elements are too high or too low. Also in draw control, the web is changing dimensionally or mechanically. ▪ Closed-loop tension control errors associated with measurement or feedback from load cell or dancer rollers, control algorithms and tuning, or integration of electronic-pneumatic-mechanical systems. Root causes specific to high tension. ▪ Excessive nip load.
Causes and Solutions to High/Low Tension with Torque Problems	1. At a steady-speed – The most obvious cause is that the tension-generating motor, brake, or clutch is oversized <i>or undersized</i>. Solution: Size torque elements properly for tensioning torque requirements. 2. <i>The torque output from the motor, brake, or clutch may be sufficient, but some non-web element is creating a torque loss (e.g. bad bearing or bushing, heavy nip load, high drag gearbox, or torque losses from other drive train element).</i> Solution: Identify and eliminate unintended torque losses. Monitor motor torque levels over time to identify changes in system performance. 3. During accel/decel – Especially with high inertia elements, such as winding and unwinding rolls (especially with high density material, such as foils) or heating and cooling rollers, inertial torques may dominate the application torque needs, creating large tension variation in systems without inertia compensation in the tension control algorithm. Inertial torque during acceleration will tend to increase downstream tension <i>and decrease upstream tension</i>. During deceleration, inertial torques will tend to decrease downstream tension <i>and increase upstream tension</i>. Solutions: Avoid aggressive accel/decel. Switch from braked to driven unwinding. Include inertia compensation algorithms in tension control programming. 4. At winding or unwinding, motor, brake, or clutch may not have a torque range capable of reaching the extreme combinations of tension, width, and diameter. Too high tension will occur when trying to run below the low end of torque range with combinations of low tension, high taper tension, narrow web, and small diameters. <i>Too low tension will occur when trying to run above the high end of torque range with combinations of high tension, low or reverse taper tension, wide web, and large diameters.</i> Solutions: Make process concessions in tension, width, and diameter ranges to keep torque range below 30:1. Require minimum taper when running highest torque combination of tension, width, and diameter. <i>Change from center winding to surface winding to eliminate roll diameter buildup ratio as torque range multiplier.</i> 5. Open-loop control errors:

	Differential winding systems usually run open-loop torque control. Errors in tension control, too high or too low, are dependent on the reliability and accuracy of the chain of events between the operator screen and the web. Working backwards from the web: - Core torque (in-lbs) is desired tension at the roll (PLI) x web width (in.) x roll radius (in.). - Applied load to create core torque is divided radial position of contact (usually core ID/2 or side contact radius) divided by slip-clutch or core-spacer friction coefficient (which will change dramatically with lubrication, heat, or material changes). - For core-grabbing (often called cam lock system), the friction created by the core-grabbing mechanism must be sufficient to transmit torque without slippage. - For radially loading shafts, required air pressure is applied load divided by contact area at the slip clutch. - For axially loading shafts, required air pressure is applied load multiplied by the air cylinder's mechanical advantage or disadvantage and divided by the effective area of the air cylinder. - Air pressure is controlled by current-to-pressure (I/P) transducer. - Correct current needs to be calculated from desired tension, including taper function (often linear or hyperbolic) with decreasing tension from core to final diameter. (To complicate matters, in a surprising number of systems, the taper algorithm is not known.) - To correctly change tension set point from core to final diameter, the system must measure or calculate diameter, typically by rewind or winding nip arm position or ultrasonic sensor. - Recipe, operator, or engineer must set the right values for start tension and tension taper. Can you spot any places where this chain of events could go wrong? (Just about anywhere, some more likely than others.) [Note: Some high-end differential shafts replace pneumatic frictional elements with magnetic hysteresis clutches, eliminating some sources of errors, but introducing others.] Solutions: Calibrate your differential winding system with a spring scale or close the loop using a load cell. Maintain differential shafts to ensure consistent performance over time. For core-dependent shafts, evaluate, specify, and inspect incoming cores to create consistent winding.
Causes and Solutions to High/Low Tension in Draw Control	Draw (speed ratio) is overly positive or negative: In draw controlled systems, carefully monitor roller surface speeds to ensure successive driven rollers do not create a positive (tensioning) that will raise tension to an undesired level <i>or a negative (relaxing) draw that would relieve all of web strain required to maintain tension.</i> Draw errors can be anywhere along the system to set roller surface speed, starting with motor speed, transmitted and multiplied (or divided) through the

drive train and gearing, and converted to surface speed through roller diameter. Roller diameter changes are the most common cause of a change in tension from a draw system. For any two draw controlled rollers, diameter decrease in the first roller or increase in the second roller will increase draw zone tension. *Draw zone tension would decrease in the opposite cases – diameter increase in the first roller or decrease in the second roller.*

The most common causes of diameter change are: a) replacing a driven roller with a new roller of the wrong diameter, b) increasing their diameter with a roller wrap, or c) decreasing their diameter through abrasion. For stiff materials, web strains may be below 0.1 percent are significant (e.g. a 0.003-in change on a 3-in diameter roller, the equivalent of one wrap of masking tape, is significant).

A lesser known source draw zone error is the effect of nipping rollers on the speed-ratio controlled rollers. A nipping roller will temporarily add strain to an elastic web at the highest friction contact point of web and roller. Increasing nip load on a draw zone driven rollers will reduce the amount of web fed in to or out of a draw zone (and decreasing nip load will increase web feed rates) with dramatic effects similar to decreasing roller diameter. For any two draw controlled rollers, nip load increase on the first roller or decrease in the second roller will increase draw zone tension. *Draw zone tension would decrease in the opposite cases – nip load decrease on the first roller or increase in the second roller.*

Some special cases of nips and draw control include: calendar roller stacks and surface winding.

Solutions: Carefully monitor motor, drive train, and roller factors that set surface speed. Take care to ensure draw controlled rollers are the right diameter. Pay attention to nip loads in draw zones.

Tension upstream of draw zone is increased or decreased:

Draw zone tension is controlled by two factors: the speed ratio of the rollers (and non-slipping web) of the speed ratio controlled driven roller and the baseline strain of the web entering the draw zone created by the upstream tension zone.

Solution: Always precede draw zones with constant tension zones, preferable with closed-loop tension control.

Web is shrinking or growing:

In a draw controlled tension zone, if the web shrinks or contracts in the machine direction after entering the draw zone (e.g. cooling or polymer free volume or stress memory recovery), the draw zone tension will increase above tension expected from elasticity only effects. In a draw controlled tension zone, if the machine direction thermal expansion exceeds the incoming web strain, the web tension will go slack.

Solutions:

To avoid high tension, decrease incoming web strain, decrease the speed ratio in the cooling draw zone, or reduce machine direction thermal contraction by reducing the temperature decrease in the draw zone by pre-chilling the web or using a web with a lower coefficient of thermal expansion.

To avoid low tension or slackness, increase incoming web strain, increase the

	<i>speed ratio in the heating draw zone, or reduce machine direction thermal expansion by reducing the temperature increase in the draw zone by pre-heating the web or using a web with a lower coefficient of thermal expansion.</i> Web is losing strength or viscoelastic: <i>Draw control tension is created by an elastic web's resistance to elongation or strain. If the web loses strength or becomes highly viscoelastic and does not resist elongation, then draw control (or any tension method) cannot create tension. Papers and tissues lose strength when wet. Thermoplastic films lose strength or become viscoelastic at elevated temperatures (or long times). Web splices may fail if paper, film, or adhesive lose strength from moisture, solvents, or heat.</i> <i>Solutions: Control moisture in papers, avoid excessive heat on thermoplastic polymer webs, avoid splice damaging conditions or design a robust splicing mechanism.</i>
Causes and Solutions to High/Low Tension with Closed-Loop Control Problems	Error in measurement or control: Advanced tension control systems are a coordinated effort of mechanical, electronic, and software systems. Closed-loop tension control systems rely on accurate measurement or response to tension changes. 1. Tension roller errors - Roller weight not zeroed properly, error in setting gain correctly for intended wrap and orientation, calibration drifts with temperature, roller threaded incorrectly, or load cell damaged. 2. Dancer roller errors - Force exerted on the web is too high or too low due to dancer system design errors, including: roller is too heavy (vertical moving dancers or standing dancers), air cylinders are wrong size, or the mechanical leverage of cylinder to roller is incorrect.
Causes and Solutions to High Tension From Excessive Nip Load	Like rolling out cookie dough, a nipping roller creates pressure that tends to reduce the thickness of a web, but also through the Poisson's effect, attempt to elongate the web in the length and thickness direction. Except for narrow and lubricated nips, the lateral stresses attempting to increase the web width are opposed by equal stresses pushing back from neighboring lanes of web. Only in the machine direction is the web free to react to the nipping load by increasing web length and tension. For many nipped systems, the thickness direction pressures and machine direction tension increases are less than 100 psi (0.1 PLI per mil of thickness). In most cases, the tension increase from nipping is elastic and will not cause yielding or breaking. However, for web with low yield points (like cookie dough), low break points, or have imperfections making them sensitive to small tension increases (nicked edges, misaligned splices) from nipping may be catastrophic. Excessive nip loads may occur when gravity and external loads are applied beyond the intended level, such as when a nip closes with additional load from inertia or when nip loads are concentrated in a local area from diameter variations, nip misalignment, or web thickness profile variations. Average nip pressures (in psi) will be higher for a given load (in PLI) in nip systems with two steel rollers, if one or both rollers have a small diameter, and thinner webs (since thick webs can act like compliant coverings).

Solutions: Use at least one compliant roller (if possible), ensure good nipping roller design (uniform diameter, uniform loading, and good alignment), minimize product thickness variations, and prevent nicked edges and misaligned splices.

1B: Tension Varies Over Time

Definition	Tension continuously oscillates above and below set point. Tension surges high or low associated with known events.
How to Measure	➤ Web tension will vary over time as measured by a load cell roller. (Note: Avoid overly filtering load cell measurement.) ➤ Web width will shift from wide to narrow. ➤ Sag in horizontal spans will increase and decrease ➤ Idler rollers will shift from turning to not turning.
Problems	✓ Coating or extrusion process variation ✓ Intermittent scratching ✓ Lateral oscillation ✓ Roll telescoping
Mechanisms And Root Causes	1. Dancer hysteresis 2. Closed-loop control errors 3. Chatter in drive train, eccentricity, and run out or bounce in roller or roll, 4. Intermittent slip on roller, esp. on driven rollers 5. Constant torque with changing diameter 6. Acceleration, steady speed, deceleration 7. Upsets during roll transfers
Solutions To Dancer System Errors	Dancers are position feedback devices that push into the web with a force to create the desired tension and move in response to imbalances in the speed of web entering and exiting the dancer loop. To function properly, dancers are constantly dithering (either obviously or to an unseen degree), constantly moving first to accumulate web when input speed is higher than output speed, then reversing to dispense web when input speed is lower than output speed. The dancer feedback is not web tension, but the position of the dancer roller, which is used to trim either input or output speeds. The goal dancer feedback in closed-loop tension control is to maintain tension at a desired set point. The total system load of the dancer exerts force on the web creating tension. In most dancers, the web wraps the roller such that two webs in the opposite direction as the dancer load and tension is one-half total dancer load. However, even the best dancers are imperfect systems, creating load variations that directly change web tension, much related to changes in load as the dancer switches constantly from accumulating to dispensing. The top sources and solutions to dancer system hysteresis are: 1. System inertia: Both translational and rotational inertias will affect dancer performance. To reduce translational inertial, use low mass rollers and components. In pivoting dancers, the components furthest from the pivot point are most important. Do not use a mass to counterbalance dancer roller weight. Counterbalance in the pneumatic system through the tension to pneumatic

	algorithm or by using a double-acting pneumatic cylinder (or second cylinder) working against gravity. To reduce rotational inertia, reduce the dancer roller inertia within design constraints (minimizing deflection). Minimize dancer mass (thin wall, light and strong materials, such as aluminum or carbon fiber). Do not add a fly wheel to the dancer (unless specifically tuning the dancer to a known frequency upset). 2. System friction: Design dancer system for minimum friction. Friction should be minimized in the mechanical and pneumatic components. Pivoting systems tend to have less friction than linear sliding system for two reasons: 1) the roller and web have a mechanical advantage over the sliding friction (at the pivot point vs. at the sliding blocks), and 2) linear sliding systems are commonly designed with undersized sled length relative to the width of the system and tend to rack (go out of alignment), leading to stick and binding friction. Only use low friction pneumatic cylinders in dancer systems. Several options are available, including rolling diaphragm cylinders with bearings on the cylinder shaft and cylinders design to lubricate there motion with air bearings. 3. Pneumatic hysteresis: The flow of air into and out of dancer cylinder needs to be of sufficient volume and unrestricted. Especially with dancers that require large volume air cylinders, either large diameter or long strokes, the pneumatic system needs to have enough flow to 4. Poor geometry: The wrap angle of the dancer roller should have minimal change as through the dancer's full accumulating and dispensing motions (usually a 180-degree wrap parallel to the roller motion at the null point.) Also, avoid a 'standing' dancer – a pivoting dancer where the pivot point is below the roller. A standing dancer is inherently unstable with gravity working to pull the roller off the null position. At low tensions, the dancer may flop to one side and not come back. 5. Position sensor: The dancer position is measured with a position sensor, either measuring the roller position direction or rotation of the dancer pivot. This motion should be detected with high accuracy and minimum backlash. Any looseness, slip, or lack of resolution in position sensor will greatly degrade the closed-loop control. In many systems, the subtle roller angle changes are magnified with a high ratio gear or timing belt system, but make sure this is a 'tight' system.
Solutions To Closed-Loop Control Errors:	1. PID loop tuned for one web or web path, but works poorly for alternate webs or web paths. (e.g. thick vs. thin web, tension through vs. bypassing air floatation oven, long vs. short web path, high vs. low density roll inertia). 2. Lack of inertial compensation. Without inertia compensations, tensions will increase at unwinds during acceleration and winders during deceleration. <i>Tension will decrease at unwinds during deceleration and winders during acceleration.</i> 3. In closed-loop speed trimmed tension control, either lack of line speed reference or trim percent is set too high or low. 4. Pairing motors and drives of differing performance. If a sluggish motor is

	paired with responsive ones, during acceleration the sluggish motor will lag the coordinate speed change <i>decreasing upstream tension</i> and increasing downstream tension.
Solutions To Other Tension Vs. Time Errors:	1. Chatter in drive train, eccentricity, and runout or bounce in roller or roll. Solutions: Good alignment and stiffness in drive shafts, zero backlash coupling, ensuring tension change to hold gear teeth meshing to one side, flat belts over timing belts, belts with sufficient stiffness, simplify drive train (build motor on shaft, eliminate gear box). 2. Intermittent slip on roller, esp. on driven rollers The coefficient of friction is a function of web and roller surface material and roughness. A roller may change its contribution to the traction coefficient over time by aging effects (like rubber hardening or oxidation) or surface contamination. The web can change its contribution to the traction coefficient at product changes, splices, on vs. off coating. Traction can also change with bagginess interaction with air lubrication. 3. Constant torque with changing diameter Many inexpensive unwinders have only a constant torque brake coupled to the roll core controlling, leaving the operator responsible for reducing brake torque as the roll diameter decreases. 4. Acceleration, steady speed, deceleration The effects of roller inertia during acceleration and deceleration will be covered extensively in the next section on tension variations within a zone. 5. Upsets during roller transfers All roll transfer methods create tension upsets. Zero-speed splicing, whether manual or automatic, have tension upsets associated with deceleration and acceleration to stop the winder or unwind and return to process speed. At-speed splicing will have tension upsets associate with speed changes required during turret indexing (increase at index start, decrease at end of index), timing problems of switching tension control from one spindle to another, delay in tension transfer to the new core, delay in severing the expired roll, change in tapering tension set point from final roll to new core value, and accurate diameter information critical for speed matching.

1C: Tension Varies Upstream or Downstream Within a Zone

Definition	Within a tension zone, the entire web from drive point to drive point is typically considered to be running at one tension set point. However, as the length of web or the number of rollers in a zone increase, there are many factors that may cause a significant change in tension as the web moves from upstream to downstream within the zone.
How to Measure	➤ Install two load cell rollers in one zone – one at the beginning and one at the end of the zone. The differential of the tension readings will show the total MD tension variations in the zone. ➤ Web width will shift from wide to narrow or narrow to wide as it passes through a zone. ➤ Identical idler rollers with identical wrap angles will turn in one end of a tension zone, but not the other.
Problems	✓ See “Problems” listed above under Tension Too High, Too Low, or Slack.
Mechanisms And Root Causes	Many of these causes are friction or traction related: bearing drag, air drag, or contact with sliding elements. Other causes of tension changes come from other applied forces: inertial torques of rolls or roller, gravity, mechanical bending losses, and roller deflection losses. ▪ Idler torque from drag and inertia is too high ▪ Web-roller peel force, air drag, or bending losses are too high ▪ Transport roller is intentionally clutched or braked ▪ Excessive cascading draw ▪ Catenary sag, including weight of coating ▪ Gravity in vertical spans, including weight of coating
Causes And Solutions To Tension Variations Within A Tension Zone: Idler Drag And Inertia	Idler drag – Reduce bearing drag with fewer rollers, smaller diameter bearings, non-contact seals, oil or liquid grease lubricant (eliminate Zerk fittings), proper assembly to avoid side loading, proper sizing of elements or spherical bushing/bearings to avoid excessive torque loads, lateral freedom to avoid thermal expansion loads. Idler inertia – Reduce inertia with fewer rollers, smaller roller diameters, and lower mass rollers (minimize diameter, wall thickness, and width). Reduce inertial torque by reducing acceleration and deceleration rates. Change from accumulators and zero-speed roll transfers to at-speed roll transfers.
Causes And Solutions To Tension Variations Within A Tension Zone: Peel Forces	Idler roller peel force – Cover or coat roller surfaces in contact with adhering webs with low peel force surfaces, including: high energy materials (plasma sputtered surface, Teflon or silicone materials), low contact area (sand-blasted, knurled, plasma-sputtered), or chill the roller surface. Peel force may also go down with low web-to-roller pressure or contact time; therefore, reduce tension, wrap angle, line speed, and increase roller diameter.
Causes And Solutions To Tension Variations	Air drag – Air drag decreases directly with speed and total exposed length (and viscosity, so it would be much greater in liquid than in air). Solution: As process speed increases, add more drive points to the process to minimize air drag within

Within A Tension Zone: Air Drag	any one zone.
Causes And Solutions To Tension Variations Within A Tension Zone: Bending Losses	Bending losses – Bending losses go down with lower modulus, thinner webs (thickness cubed), and increasing diameter (inverse square of radius). Solution: Reduce number of rollers or increase roller diameter.
Causes And Solutions To Tension Variations Within A Tension Zone: Clutched And Braked Rollers	Transport roller is intentionally clutched or braked – Any roller that is intentionally clutched or braked, such as tendency drive rollers in ovens, will apply drag or power to the web depending on whether the input shaft speed is above or below the RPMs needed to for the roller to reach the web's speed. Solution: Ensure clutched or braking elements are low drag or alternate between clutching and braking with over and under speed shaft RPMs of consecutive rollers.
Causes And Solutions To Tension Variations Within A Tension Zone: Cascading Draw	Excessive cascading draw – Many equipment builders believe that in speed controlled processes (such as multiple point printing or the driven rollers in the driven section of a slitter-rewinder) the speed must increase through the process. For elastic webs, this means increasing strain and tension through the process. Solution: Avoid increasing draw through a system handling elastic webs.
Causes And Solutions To Tension Variations Within A Tension Zone: Catenary Sag And Gravity	Catenary sag – A web in a horizontal span will sag due to gravity. The tension change through the sagging span, lower at the end and highest at either end of the spans. For many webs, where tension is much higher than web weight, this difference is significant. Solution: Keep tension high relative to span weight or keep horizontal spans short. Gravity – In vertical spans, the web tension will change through the webs increasing elevation, lowest at the bottom of a vertical span and highest at the top with a total difference equal to the weight of the web in that span. Solution: Keep tension high relative to span weight or keep vertical spans short.

1D: Tension Varies Across the Web's Width

Definition	Machine direction tension is carried unevenly across the web width. (Not all crossweb positions carry the same tension in PLI.)
How to Measure	➤ Low tension lanes are visibly loose, buckling ahead of rollers and nip points, sagging more in horizontal spans, or fluttering. ➤ Several options are available to measure crossweb tension variations, but few are used in production.
Problems	✓ Web breaks or yields when edge stress is too high. ✓ Wrinkles on rollers (especially equipment-induced one-sided tension variations or loose center baggy web in long spans). ✓ Wrinkles at nips (if portions of the web are at or near zero tension). ✓ Flutter (in portions of the web are at zero or low tension).
Mechanisms And Root Causes	Crossweb variations can be quickly divided between web, equipment, and splice by noting where the tension variations occur. If the crossweb tension variations are seen only in specific areas of the equipment and are there repeatedly after product or roll changes, the equipment is likely the cause. If the crossweb tension variations are seen through the web line, but change throughout the run of a given roll or after roll changes, then the web is likely the cause. If the crossweb tension variations are seen only when the splice travels through the equipment and goes away after the splice has passed by, then blame the splice. 1. Equipment error: alignment, deflection, diameter variation. 2. Variation in paper, film, or foil making process, including high thickness lanes distorting the web in wound roll. 3. Misaligned splice.
Solutions To Equipment Errors	1. Ensure equipment is level and tram to tolerance required by handling, yield, and break strain of the web. The most common alignment problems include: ▪ Initial installation errors ▪ Failure to pin or bolt equipment to floor ▪ Foundation and earth settling ▪ Unpinned roller mounts ▪ Asymmetrical deflection under load ▪ Cantilevered rollers with undersized support shaft or beam ▪ Off-center web and tension on rollers ▪ Web guides ▪ Pivoting rollers or rolls (nip roller, dancers, turret winder, or shaft arms) ▪ Linearly sliding rollers or shafts ▪ Air nozzles and air turns, especially 45-degree turn bars ▪ Asymmetrical thermal expansion

	2. Ensure roller deflection (both from roller weight and web tension) is minimized based on web sensitivity. The most common deflection causes are the response of the end-supported roller to forces applied to the roller center, including: ▪ Gravity and roller mass ▪ Nip loads ▪ Tensions (two per roller, one per unwind or winding roll) ▪ Bowed rollers 3. Ensure roller diameter variations are less than strain of the web under nominal tension. The most common causes of diameter variations include: ▪ Poor initial machining ▪ Contamination on the roller surface ▪ Uneven tape wraps ▪ Rubber covering wear ▪ Crossweb thickness variations in coating, printing, laminate width, or laminate bubbles ▪ Long-term, non-oscillated, crossweb thickness variations and winding ▪ Uneven supply or excessive side leakage on air turn
Solutions To Variations In Paper, Film, And Foil Manufacturing, Including High Thickness Lanes In Wound Rolls	This is a topic beyond web handling process technology alone. This topic is directly related to the discussion in these notes on camber and bagginess. To reduce distortion of thickness variations in winding: 1. Minimize thickness variations 2. Reduce product stack modulus through material properties (soft coating, increase roughness, soft interleave web) or wound roll tightness (low winding tension and nip load, small roll buildup, large cores). 3. Oscillation during winding: a) oscillate core or automatic guide edge sensor to wind a stagger edge roll, b) oscillate trim knives and guide to straight roll edge roll, or c) oscillate the web into fixed trim knives.
Solution To Misaligned Splices	For zero-speed manual splices, install and use a splicing table with clamps and reference lines. For zero-speed automatic splices, ensure the web is aligned in setting up the incoming web properly. For automatic splices, ensure the pasting nip roller is aligned and nipping properly.

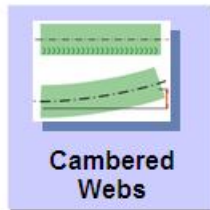
1E: Tension Varies Across Slit Strands

Definition	Tension is carried unevenly between strands after slitting. (Not all crossweb post-slitting strands carry the same tension in PLI.)
How to Measure	➤ Low tension strands are visibly loose, sagging more in horizontal spans, fluttering, and shifting laterally. ➤ In advanced systems, usually with dedicated slit widths, each slit strand will have an individual load cell roller. This approach is rarely done due to either the high cost of multiple load cell rollers, the productivity loss of changing rollers with slit width changes, or the lack of justification where no problem exists.
Problems	✓ Poor slitting quality where tension is low or zero. ✓ Lateral shifting of strands from low tension. ✓ Slit roll variations, both too tight and too loose.
Mechanisms And Root Causes	Crossweb slit strand tension variations can be divided between three main causes. 1. Input web quality (bagginess or camber). 2. Lock shaft tension changes and roll-to-roll diameter variations 3. Slip shaft tension changes from roll-to-roll torque variations.
Solutions To Input Web Quality Problems	<i>Low tension from slitting baggy web: A floppy, baggy, or slack edge or lane in a wide web may cause some problems, but as long as some of the web is under tension, it will continue to track in the machine direction and peel safely off rollers (though it may wrinkle on rollers at nip points). However, if your web has baggy lanes or edges into and out of narrow width slitting, the slit strands cut from the baggy lanes will be fully loose or slack after slitting.</i> <i>Solutions:</i> 1. <i>Increase tension before and after slitting to pull short non-baggy lanes of material to be equal or longer than the length of the baggy lanes, creating tension on all slit strands.</i> 2. <i>Use differential winding to accumulate the excess length of baggy lane slit strands. (Note: Differential winding may not resolve this problem if there are too many high traction rollers between slitting and the differential shaft.)</i>
Solutions To Lock Shaft Winding Problems	Lock shaft winding is attempts to wind multiple slit rolls locked to a common shaft all turning at the same revolutions per minute (as opposed to differential shaft winding where each roller can seek its own RPM rate). A web with no bagginess, but with small crossweb thickness variations will quickly create strand-to-strand tension variation in lock shaft winding if the thickness variations turn in to slit roll diameter variations. Especially in products with high stiffness in the thickness direction, a building slit rolls diameter will be directly related to thickness. At the start of winding, the problem is not obvious since all rolls will start on cores of the same diameter. As the set of slit rolls build, the rolls winding thicker than average web will build to above average diameter and <i>rolls winding thinner than average web will build below average diameter</i>. Above average diameter

	rolls will pull a web at a higher surface speed, creating an above average draw ratio and increased tension. <i>This added tension is stolen from the below average diameter rolls.</i> Solutions: Wind small rolls, reduce thickness direction stiffness, or use differential winding.
Solutions To Slip Shaft Winding Problems	Slip shaft (a.k.a. differential) winding is designed to prevent the most common problem of tension variations after slitting, but poor design or performance in differential shaft winding can cause equal or greater strand-to-strand tension variations. 1. Roll-to-roll clutching torque variations – Many differential shafts use fairly simple clutching mechanisms due to cost and space limits. Core-dependent differential shafts, where friction generated against the core’s inner diameter or side, are highly susceptible to core material, moisture, geometry, and edge quality variations. 2. Cores compression locking onto shaft – High tension winding exerts high pressure onto a roll’s core. With enough pressure, a thin-wall paper core may compress enough that the core is squeezing down on the shaft either increasing the shaft to core torque transmission or locking to the core. 3. Over speed supplied by weak clutch – In most modern differential slitter-rewinders, the differential shafts have an independent motor to control shaft over speed. In some cost-saving designs, the differential shaft over speed is driven off the slitter’s main drive motor via a clutch. In lock bar winding, this clutch sets the winding torque and tension. In differential winding, this shaft driving clutch needs to be set at maximum to allow the differential shaft clutching mechanism to set winding torque and tension. If not, the drive train clutch sets the tension and the differential shaft becomes an expensive lock shaft. 4. Torque from nip load – As speed increases, smooth non-porous webs need winding nips, in addition to center torque from the differential shaft, to manage air entrainment and avoid shifted layers. However, similar to gravity, any nip load applied to the winding roll will increase the force between core and shaft and increase winding tension from core-shaft friction. When using a common winding nip (one wide nip roller contacting multiple slit rolls), the nip load will vary from roll to roll due to misalignment and roll diameter variations. As nip load varies roll to roll, so will the combined slit roll tightness from center torque and nip-induced tension. 5. Torque from roll weight – Friction-based differential shafts apply a radial or axial load to create torque and tension; however, as roll size increases, the friction, torque, and tension from gravity needs to be calculated and included in the differential control algorithm. If your differential machine controls don’t ask for your roll density, then they either don’t have gravity compensation in the algorithm or assume all products have the same density –either has potential for actual winding tensions to be far from set point, usually higher.

II. Solutions to Imperfect Webs Problems

Cambered and Baggy Webs



Non-Flat Webs



Minimizing Waste from Imperfect Webs

Note: *These recommendations do not reduce the camber, bagginess, or curl in your product but aim to handling the imperfect webs without waste.*

All webs have some level of camber, bagginess, and curl.

- Low levels of camber, bagginess, and curl run without problem.
- Moderate levels of camber, bagginess, and curl will create waste.
- High levels of camber, bagginess, and curl will shut down a process.

The goal of good web handling practices is to reduce waste from imperfect webs by raising the level of camber, bagginess, or curl where waste begins.

Solutions to imperfect webs have two approaches:

1. Reduce the level of imperfection
2. Reduce the process sensitivity to imperfection

Both approaches should be pursued.

2B: Baggy Webs

Definition	A web with crossweb length variations. At zero or low tension baggy webs will not lay flat on a tabletop or will have variations in sag or droop in spans with a horizontal component. With sufficient tension to strain the shortest lanes equal to the longest lanes, a baggy web will appear uniform, but will carry tension non-uniformly, with the short and long lanes carrying above and below average tension, respectively. Baggy webs become a problem when acceptable tensions are not sufficient to strain the short lanes equal to the long lanes. Longer edge lanes are defined as baggy edge web. Longer center lanes are defined as baggy center web.
How to Measure	➤ There are many methods to qualify or quantify bagginess. ➤ Most are difficult, expensive, or time-consuming. ➤ A manual pull out and 1-to-5 grading may be the simplest. ➤ For films, roll hardness can be a useful predictor or bagginess.
Problems	Top Problems: ✓ Wrinkling at most nipped rollers and winding Nips are needed for many value-adding processes, such as coating, printing, laminating, embossing, and calendaring. However, nips are often used unnecessarily to create high friction at driven rollers. Unlike unnipped and vacuum-assisted unnipped driven rollers, nipped driven rollers do not forgive even low levels of crossweb length variations (camber and bagginess). In a uniform nip roller system, all crossweb lanes are web are feeding at a given rate. When an obviously baggy web meets a nip, the short lanes will pass through the nip without problem, but the longer lanes of web will find the nip rejects there excess length, accumulating it upstream of the nip point. The rejected length accumulation can only last for a short time before leading to a shear or accumulation wrinkle. ✓ Wrinkles of lateral shifting in long spans ✓ Lateral shifting in air floatation ovens and air turns ✓ Wound roll tightness variations ✓ Lay-flat or laminate curl complaints from customers ✓ Lateral shifting in air floatation ovens and on air turns Secondary Problems ✓ Coating variations in tension dependent coating methods ✓ Backside corona treatment ✓ Post-slitting lateral motion of slit strands ✓ Crossweb registration errors
Mechanisms and Root Causes	Top Causes: 1. Within a wound roll, the web permanently yields in high stress bands of above average thickness lanes or relaxes in low stress bands of below average thickness lanes, especially over long storage time. Secondary Causes:

	2. Paper-making process variations 3. Film-making process variations – die lip, hot spots (oven, die, calendar), calendar nip pressure variations 4. Excessive web path variations (misaligned rollers, web spreaders)
Solutions to Reduce Bagginess	Regarding the top two causes, both associated with crossweb thickness variations and winding: ▪ Reduce thickness variations ▪ Reduce roll tightness ▪ Lower tension, increase tension taper ▪ Reduce stack modulus (increase roughness, wind with soft interleave) ▪ Reduce roll buildup ratio (increase core size) ▪ Control roll aging – relax polymer on-line, relax polymer in roll form and rewind, convert quickly. ▪ Knurl edges, shifting stress and distortion to Edges ▪ Thickness oscillation at winding ▪ Decrease product stack modulus Most important: Measure Bagginess As a producer of baggy web, measure bagginess and work on process improvements incrementally. In many products, measuring crossweb roll hardness may be the first and simplest predictor of bagginess severity. As a customer of baggy webs, measure bagginess and document the increase in waste associated with increasing levels of bagginess. Work with the supplier to improve quality. Measure crossweb roll hardness to see if it is a predictor of bagginess – or more importantly, performance and waste in your process. If yes, use roll hardness as a quality acceptance tool.
Solutions for Reducing Bagginess-Related Waste	These solutions will help with both baggy and cambered webs. 1. Increase Tension – Webs are often not considered baggy if they appear flat under nominal process tensions. If the percent length variation causing bagginess or camber is less than the average strain from web tension then normal tensions will easily stretch the short lanes equal to the long lanes and run flat through the process. If a web has a slight amount of visible bagginess at normal tensions, the first thing to try is to increase tension and see if the bagginess disappears. There are limits to how high you can turn up tension without running into high tension defects (see Section 1A for common high tension problems). <i>In some cases, how much you can increase tension is artificially low due to an equipment limitation, such as slip on drive rollers or when handling tension is also winding tension. In these cases, an equipment upgrade of modifying or adding a driven roller is required before the sufficient tension to forgive higher levels of camber and bagginess can be achieved.</i> 2. Eliminate Nips (where possible) –Where possible, change nipped driven rollers to unnipped or vacuum-assisted driven rollers. Reduce driven roller tension differentials to less than 2:1 or 3:1 and increase coefficient of traction to driven rollers so they will not slip in unnipped operation.) Reduce

nip load or lubricate nips to allow long edge to slip through.

Similar to reducing baggy web problems with higher tension, in some cases, whether you can eliminate or open a drive roller nip in your process is artificially limited due to an equipment or process limitations. Under one set of process tensions, where the tension differential is high, opening the nip will cause slip or lack of control. In these cases, a change to more balanced tensions between zones or modifying equipment may be needed. Unnipped drive rollers need to have good friction to the web, good roughness is running at high speed-to-tension ratios, and good wrap angle.

3. Apply Spreader Rollers (especially upstream of nips and winding) – Use just upstream of nipped processes. Best spreaders for baggy webs: bowed roller, edge nips, expanding surface roller, D-bar.
4. Use a Carrier Web or Support Belt for Tension-Sensitive Coating Processes – In fluid bearing coating, uneven tension of baggy webs may create uneven coating thickness. Avoid problems of uneven web tension by using a uniform carrier web or belt to provide coating process tension.
5. Shorter Spans (More Rollers) – Designing new equipment with short spans or adding more rollers to existing equipment to create short spans and will reduce the ability of baggy web to gather or fold over on itself. Many camber and baggy web problems are caused by the web tracking to the left, right, or center. Long spans allow an imperfect web to move further laterally and require less driving force to do so.
6. Air Impingement Optimization – Force air impingement, such as with aggressive dryer nozzles and air turns, will create a lateral force on baggy or cambered webs that favor higher tension on one side over another. The lower tension side of the web will float higher off the air supply surface creating a crossweb angle in the web. Like how wind catches a sail and drives a boat, the force of high velocity air will drive a web off centerline.

Solution: Reduce air velocity, amplitude, and lengths between roller contact in air floatation ovens. Reduce span lengths into and out of air turns. Design air turns with roller on the input side. Use air turn side flanges or air hole patterns to discourage lateral motion.

2B: Cambered Webs

Definition	Camber is the most common term for a non-straight web. Camber is typically measured by pulling, rolling, or sweeping out a long sample on a tabletop or floor and quantifies the left or right bias of the web relative to straightness. Most commonly, camber is the total deviation, left or right, off a straight line. Skew is an alternate term for this measurement. Alternately, camber measured after folding the sample width-wise aligned at both ends and measuring the lateral misalignment at the mid-way point, essentially measuring half the bow of the sample. Camber is usually used to describe slit rolls rather than mill or jumbo rolls. In a paper, film, or foil making process, as the wide jumbo or mill roll is slit down to narrower widths, camber is usually low for the center cuts and progressively worse for edge cuts.
How to Measure	➤ Camber is typically measured by pulling, rolling, or sweeping out a long sample on a tabletop or floor and quantifies the left or right bias of the web relative to straightness.
Problems	Many of the same problems as baggy webs. 1. Wrinkling or web breaks at most nipped rollers, blade coaters, extrusion nips, web guide sensors, and winding 2. Wrinkles or lateral shifting in long spans 3. A cambered web will form shear wrinkles with less bending if the equipment error pulls to the cambered web's loose side. 4. Lateral shifting in air floatation ovens and air turns 5. Wound roll tightness variations 6. Lay-flat or laminate curl complaints from customers
Mechanisms and Root Causes	Camber is usually used to describe slit rolls rather than mill or jumbo rolls. In a paper, film, or foil making process, as the wide jumbo or mill roll is slit down to narrower widths, camber is usually low for the center cuts and progressively worse for edge cuts. Machine misalignment, crossweb moisture variations, variations web paths or nip loads in blown film bubble collapsing, cooling or stress variations in tenter quenching zone, nip variations in laminating or film extrusion nips, uneven calendaring. In narrow slit rolls, camber represents a sampling of bagginess in the input web. If a slit roll is cut from a baggy web across a lane in the transition from short to long lanes, the slit roll will be cambered.
Solutions to Reducing Camber in Webs	Solving paper, film, and foil making process variations is beyond web handling alone, but web handling must always consider how to make acceptable product from marginal quality input materials. 1. If camber is from slitting baggy web, see the previous section for solutions to reduce web bagginess of web supplied to slitting. 2. Request Center Cuts – Often there will be a strong trend from the suppliers of camber by cut position. In paper or film making where four cuts are made

	from a wide process, the center cuts will have low camber and the right and left most cuts will lean in a characteristic direction.
Solutions to Reduce Process Sensitivity to Cambered Webs	In addition to all the solutions listed to reduce sensitivity to baggy webs (see previous section), these actions can reduce sensitivity specifically to cambered webs. 1. Run Similarly Cambered Webs Consecutively – For cambered webs, operators will learn how to make slight adjustments in nip load, spreader rollers, tension, and adjustable rollers. To maximize the benefits of these adjustments and reduce re-adjustments, group the rolls of known cambered webs and run all the left-tending rolls as a set separate from the right-tending rolls. Ask your supplier to package and palletize the rolls to group the right cuts and left cuts so they can more easily be staged and ran as a set. 2. Use Adjustable Roller – Ahead of nipped processes, an skewing or tramming roller can tighten one side of the web preventing an accumulation wrinkle. Changing the web path length with a manually skewed roller can improve even tension distribution. (Note: I have never seen a successful automated system in production, though I have seen these devices claiming they do so at trade shows.)

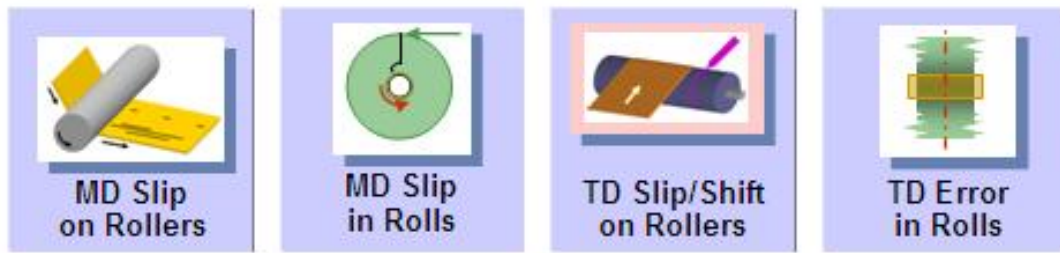
2C: Curled Webs

Definition	Lay-flat is the comparison of a web sheet sample relative to a flat surface such as a tabletop or sheet of glass. Lay-flat problems come in many shapes and degrees, including puckered areas, rippled edges, and curl. Curl is a tendency for the web to not lay flat, instead forming into a curved or scroll shape. Curl will usually form in a preferred direction either MD or TD and sometimes diagonally relative to MD and TD. Curl is best measured in narrow strip samples. Strips samples are usually cut in the machine or transverse (crossweb) direction, but may be cut at any angle.
How to Measure	➤ Curl is best measured in narrow strip samples. Strips samples are usually cut in the machine or transverse (crossweb) direction, but may be cut at any angle.
Problems	✓ Curled web may fold over in handling, especially in long spans. ✓ Customer requiring untensioned flatness in their end use.
Mechanisms and Root Causes	Curl is created by mismatched in dimensions across the thickness of a web, usually found in coated or laminated webs of two or more bonded layers, but also found in single ply products. Dimensional changes are created by stress, thermal expansion, hygroscopic expansion, polymer relaxation, viscoelastic flow, and yielding. 1. Strain Mismatch – Stress differences in laminates are most commonly caused by tension of webs at combining, but may also be induced by speed variations in when both laminator rollers are driven or from non-straight exit from the laminator nip. 2. Thermal Expansion or Contraction – Thermal-induced curl occurs when two webs of differing thermal expansion are joined at one temperature (usually elevated from room temperature), then cooled to differing dimensions with the product curling toward the layers or layers that have higher coefficient of thermal expansion. 3. Shrinkage – Shrinkage-induced curl occurs if a coated layer bonds to a web during initial drying but continues to contract with additional drying or curing. 4. Moisture – Moisture-induced curl is similar to thermal-induced curl, except it occurs in coated or laminated webs with differing coefficients of hygroscopic expansion if when the moisture content change causes a dimensional mismatch. Papers and nylon films have much higher coefficients of hygroscopic expansion than most films and foils. If a paper-film web is flat under a low humidity environment, it may become curled when put into a high humidity environment. As the laminate absorbs moisture, the paper will expand, especially in the crossweb direction, making the laminate curl away from the paper. Papers can curl from hygroscopic expansion due to through-thickness moisture variations, especially one-side poly-coated papers. 5. Relaxation – Polymer relaxation is a dimensional change that is often related to thermal processing, but it is not directly related to temperature change. Like viscoelasticity is a time-dependent reaction to stress, polymer relaxation is a time-delayed reaction to processes with changing temperatures and

	stresses. 6. Yielding or Viscoelasticity – A single layer web may curl due to yielding or viscoelastic flow when the product is either stressed through small radius curvature or stored in a curved shape over long times, such as within a wound roll.
Solutions To Web Curl Root Causes	1. Strain Mismatch – Set tension of webs proportional to their spring constant (thickness x width x elastic modulus) to match strains at laminating. Avoid excessive pre-laminating wraps on rubber rollers. Avoid excessive laminating nip loading. Design web path to exit straight out from the nip point, perpendicular to the nip loading direction. 2. Thermal Expansion or Contraction – Calculate the dimensional change from thermal expansion or contraction and minimize differential during any elevated to room temperature transition. Example: Hot laminating heat-sealable PET to copper foil. 3. Shrinkage and Relaxation– MD shrinkage may be a problem when tension is removed, but usually doesn't cause problems in handling. TD shrinkage of a coating will promote crossweb curl, leading to folded edges when the web isn't stiff enough to resist the shrinkage stress, especially with long spans and low tension. As noted above, short spans and induced MD curvature can postpone problems with TD curl. Avoid combining freshly made polymer films without delayed storage or heat processing. Avoid handling polyester films above its glass transition temperature. 4. Moisture Change – For paper products, measure and control moisture content prior to laminating. For paper-film laminates, consider use of nylon in place of non-moisture absorbing polymers. Example: Laminating paper to paper, PE, or Nylon. 5. Yielding – Prevent yielding-induced curl by not bending beyond its critical radius of curvature. The critical radius of curvature is a function of yield stress, modulus, and thickness. Solution: All rollers and cores should be larger than the critical radius of curvature. 6. Viscoelasticity – Prevent viscoelastic curl by avoiding the combination of materials with strong viscoelastic behavior, small radius of curvature (which creates the stress that drives the viscoelastic strain), long times (which allow the viscoelastic strain to become significant), and environmental conditions that promote viscoelasticity. Solution: Wind on sufficiently large cores and avoiding long or hot storage conditions. 7. Laminate Product Symmetry – A laminate product with an A-B-A construction may uneven strains between layers, but if the A layer stresses are equal and opposite to each, the laminate will remain flat. 8. Increase Product Stiffness – Curling is a bending. Thicker, higher modulus webs will bend and curl less than a thin, lower modulus web under the same stresses.
Solutions to Reduce Process Sensitivity to Crossweb Curled Webs	1. Shorter Spans (More Rollers) – A running web with a strong crossweb curl may want to fold over on the rollers at the end of long spans. Designing new equipment with short spans or adding more rollers to existing equipment to create short spans and will reduce the ability of curled web to fold over.

2. Support Web with Rollers on the Inside of Curl – As a curled web approaches a roller it will likely behave differently when it approaches a roller with the web curling towards or away from the roller contact. In some cases there is a dramatic difference where contacting the crossweb curling web with rollers on the inside of the curl. The roller will have a tendency to keep the crossweb curling web ‘open.’
3. Induce MD Curvature – Webs often benefit from shape stiffening. It is usually difficult for a web to bend in two planes at once. A crossweb curled web will typically lay flat when curved in the machine direction while wrapping a roller. Any curvature in the machined direction can help restrain crossweb curl from folding over. Options for inducing MD curvature include: rollers, drag bars, air turns and air bars, vacuum-created catenary droop (where the web is pulled out of plane between two rollers with a pressure differential).

III. Solutions to Slipping and Shifting Webs Problems



3A: MD Slip on Driven or Idler Rollers

Definition	The web runs faster or slower than the surface speed of a driven or idler roller.
How to Measure	➤ For driven rollers, if one drive roller runs faster or slower than other drive roller by a percent outside the reasonable strain of the web it must be slipping. ➤ Idler roller slip is rarely measured, but a simple handheld contact tachometer can be used to check for web vs. roller speed differentials of greater than one percent. ➤ More advanced systems use targets and optical tachometers to monitor idler roller speeds.
Problems	✓ Scratching, abrasion (of web or roller) ✓ Loss of speed control (causing coating variations in flow metered coating methods and many other length / speed related defects) ✓ Loss of tension control (see first section for problems of poor tension control) ✓ Loss of parallel entry rule: ○ Loss of centerline tracking benefit of cambered webs on aligned transport rollers. ○ Loss of control in steering and displacement type web guides. ○ Loss of outward displacement mechanism on good traction, diameter-based spreaders (tape collar roller and concave rollers), leading to gathering or wrinkling. ○ Loss of outward displacement mechanism on good traction, misalignment based spreaders (bowed, flat expander, and flex expander rollers).
Mechanisms And Root Causes	The tension change across a roller exceeds the traction available. Traction Safety Factor is less than one. The TSF is the traction available divided by the traction demand. Traction Available Unnipped Roller. The traction on an unnipped roller is a function of coefficient of traction, web tension (in force per width), web width, and wrap angle. The lateral friction available is a simple product of these factors, but the machine direction

traction is governed by the belt equation. The belt equation takes into account how machine direction sliding friction from micro-slip between web and roller will alter the tension as a function of wrap, making the traction available increase or decrease from the non-slip value. The belt equation does not enable any absolute traction level, but states the traction limit as a ratio of incoming to outgoing tension.

The coefficient of traction is the same as the coefficient of friction when air entrainment is insignificant. As the entrained air layer grows to a height similar to the combined roughness of the web and roller, the web-roller interface will begin to lubricate. The 'apparent' coefficient of friction, now called the coefficient of traction, will continue to be proportional to increases in the entrained air layer height until the web-roller interface is swimming in air and the coefficient of traction is essentially zero.

Nipped Roller. The traction created by two rollers nipping the web is a function of the web-to-roller coefficient of friction and the total load exerted on the web. There are three different COFs that may determine the friction limit in a nipped roller system: the COFs of web to roller A, web to roller B, and roller A to roller B. In the case of a thick web where the nipped rollers do not make contact outside the web's width, the roller to roller COF does not apply.

In the case where the web is thin and one or both of the rollers conforms or deflects around the web to make contact outside the web width, then the roller to roller COF may be important. For the web to slip in this thin web case, two of the three contacting combinations must slip. Either the web must slip relative to both rollers or the web and one of the rollers must slip relative to the other roller.

Vacuum Roller. For a vacuum roller (a.k.a. suction roller), the friction created from tension over a wrapped roller is supplemented with the friction created by the pressure drop. Friction from the suction roller pressure drop will increase with web-roller coefficient of friction, the belt effect of the increasing wrap angle, and increase normal load created from the effective pressure drop across the web and the total effective area exposed to pressure drop.

Note: The effective pressure drop is the difference from atmospheric pressure and lower pressure between the web and roller, not the suction roller's internal pressure. This difference is dependent on the pressure loss through the roller's shell, which is a function of suction roller design. Also, the effective area exposed to the pressure drop will increase with web width and wrap angle, but does not equal the total roller contact area. The web only feels the suction roller pressure drop in the area immediately near the hole or groove pattern.

Traction Demand

The traction demands of an idler roller are covered in the 'Tension Varies Within a Zone' discussion regarding idler roller drag and inertia. In this case, instead of viewing drag and inertia from a series of rollers, consider the drag and inertia from a single roller. The solutions to reduce tension losses from drag inertia apply here as solutions to keep the traction demand for a roller below the traction available.

To determine the traction demands of a driven roller, first consider the nominal tension differential created by the tensioning systems in the upstream and

	downstream tension zones. In the case of closed-loop tension control, the nominal tension is the value measured by the load cell roller or created by the dancer roller pushing into the web (with all the potential errors in these methods listed above). In the case of torque controlled tensioning, the nominal tension is the applied torque divided by the roll or roller radius (with all the potential errors of torque system listed above). In the case of draw control, the nominal tension is set by the baseline strain of the web entering the draw zone and the change in strain created by the positive, neutral, or negative draw (with all the potential errors of draw control systems). To find the traction demand across a driven roller, start with the nominal upstream and downstream tension (and the errors listed in the previous paragraph) and add to this nominal zone-to-zone differential the additional tension change within the upstream and downstream zone due to all the factors included in the 'Tension Varies Within a Zone' discussion. Traction Safety Factor will drop below one due to: <u>Traction available</u> is too low: 1. Tension-induced friction (non-lubricated) decreases with lower web-roller coefficient of friction, wrap angle, and tension. 2. Nip roller loads or effective vacuum roller pressure and area are too low. 3. Entrained air exceeding combined web and roller roughness, texture, or porosity. <u>Traction demand</u> is too high: 1. Traction requirement increase with increasing torque from poor roller performance and inertia during acceleration or deceleration. 2. Traction requirement on drive rollers increase with tension changes from zone to zone.
Solutions To Low Tension-Induced Friction	1. Increase tension in the entire tension zone (increase air to dancer roller, set point for load cell feedback, torque to clutch or brake, speed ratio). 2. Increase friction coefficient by covering or wrapping roller with high COF to web. 3. Increase tension locally by adding or removing drag or inertial losses from idler rollers within a tension zone. To increase tension on the upstream end of a tension zone, reduce roller drag. To increase tension on the downstream end of a tension zone, increase roller drag. 4. Increase wrap angle on rollers by adding idler rollers or changing the threading order of existing rollers. 5. In wrap-limited systems, remove rollers to increase wrap angle on remaining rollers. 6. Use positive or negative air pressure to increase the catenary sag between rollers and increase wrap angle. Use Coanda effect near rollers to create negative pressure and increase wrap angle.
Solutions To Low	1. Increase nip roller load or coefficient of friction between web and driven

Nip Or Vacuum Roller Friction	roller. 2. Increase roller conformity around web with larger diameter, thickness rubber covering, and lower Durometer hardness. 3. Increase load on web by minimizing deflection and width of nipping rollers outside web width. 4. Increase wrap angle, diameter, or effective vacuum exposure area of vacuum rollers.
Solutions To Entrained Air Exceeding Roughness, Texture, Or Porosity	1. Reduce entrained air with decreased radius, speed, and air viscosity and increases with tension or nip load. 2. Increase web roughness through substrate manufacturing process (slip additives, quench or calendar roller roughness) or post-processing (coating, patterned printing, embossing, scratching, knurling). 3. Increase roller roughness by higher machined roughness, sandblasting, plasma sputtering, ceramic coating (rough), or wrapping with a material of increased roughness. 4. Increase porosity of web by perforating. Increase roller surface porosity by holes, perforation, or air-absorbing wrap material (cheesecloth). 5. Reduce air lubrication locally with crossweb tension variations induced by misalignment, diameter variations, or baggy web. In each case, lanes with above average tension will maintain traction longer than cylindrical roller. 6. Reduce entrained air by processing in vacuum system.
Solutions To Poor Roller Performance And Inertia Losses	(Covered previously in tension loss within a zone, Section 1C.) 1. Idler drag – Reduce bearing drag with fewer rollers, smaller diameter bearings, non-contact seals, oil or liquid grease lubricant (eliminate Zerk fittings), proper assembly to avoid side loading, proper sizing of elements or spherical bushing/bearings to avoid excessive torque loads, lateral freedom to avoid thermal expansion loads. 2. Idler inertia – Reduce inertia with fewer rollers, smaller roller diameters, and lower mass rollers (minimize diameter, wall thickness, and width). Reduce inertial torque by reducing acceleration and deceleration rates.
Solutions To Zone-To-Zone Tension Changes	Reduce tension changes between zones. 1. Change tension set points, torques, or draws to create even tension into and out of drive rollers. 2. Reduce tension losses that create tension changes within tension zones, leading to confusion on what actual driven roller tension change is. 3. Place feedback devices near driven rollers to minimize confusion (at least on one side) of what the input or output tensions are. 4. When large tension changes are required, use multiple driven rollers to reduce the tension change over any one driven roller.

3B: MD Slip within a Winding or Wound Rolls

Definition	The torque applied between some or all layers within a wound or winding roll exceeds the traction available to transmit that torque (a.k.a. torque capacity), causing layers to slide in the direction of the applied torque. Though cinching can occur at winding or unwinding and in the tightening or loosening direction, the most common cinching in the tightening direction at unwinding, usually starting in layers near the core.
How to Measure	➤ At unwinding, mark a spoke line on the side of the roll before tensioning. If any part of the spoke line shifts, MD slip has occurred. ➤ MD slip in winding rolls is more difficult to measure. An inkjet printer synchronized with the winding roll will form a spoke line (no slip) or D-line (indicating MD slip).
Problems	✓ Scratching, abrasion ✓ Loss of tension control ✓ Roll tightness increases or decreases ✓ Cinching-induced telescoping ✓ Crepe or cigar wrinkle in paper winding
Mechanisms and Root Causes	1. Cinching from center tensioning torque transmission: A roll's torque transmission capacity is calculated throughout a roll by multiplying any layer's internal radial pressure by its area of contact and side A-to-B coefficient of friction. In center winding or unwinding, the torque is applied to the roll from its center to create tension at the roll's outer diameter. The onset of cinching related telescoping is most commonly seen as a roll grows in diameter and first appears in the layers near the core and the outer wraps. The problem occurs as center winding rolls growing diameter require more torque to create tension. The cinching onset begins near the core since these layers have less area and are at a mechanical disadvantage to the outside of the roller (creating a radius-square effect, with outer layers having more torque capacity). Since the relationship of tension to pressure is non-linear, a small increase in tension can greatly increase torque capacity. Also, a roll with flat thickness profile will have less torque capacity than a roll with gauge variations. Thick, low friction webs are the most prone to cinching-related telescoping. 2. Cinching from inertial torques: During acceleration and deceleration, inertial torque is transmitted through the roll. In center winding or unwinding, the inertial torque is opposed by the center drive. In surface winding or unwinding, the inertial torque is opposed by the surface drive. Unwinding, loosening/tightening cinch: If you aggressively accelerate a large unwinding roll from its core the outer roll layers act as a flywheel opposing this speed change. The layers near the core will accelerate faster than the roll's outer layers causing a loosening cinch. With aggressive deceleration,

	the opposite happens, tightening the roll. Center winding, tightening/loosening cinch: If you aggressively accelerate a large winding roll from its core the outer roll layers act as a flywheel opposing this speed change. The layers near the core will accelerate faster than the roll's outer layers causing a tightening cinch. With aggressive deceleration, the opposite happens, loosening the roll. Surface winding, loosening/tightening cinch: If you aggressively accelerate a large winding roll from its surface the inner roll layers act as a flywheel opposing this speed change. The layers near the core will accelerate slower than the roll's outer layers causing a loosening cinch. With aggressive deceleration, the opposite happens, tightening the roll. 3. Cinching from core tightening cinching: Core tightening cinching can be a defect with a viscous cycle where the onset of the defect creates more of the defect. If unwind center tensioning cinching causes the layers near the core to tighten, these layers will move to a small radial position, falling away from the outer layers of the roll. This radial contraction reduces the pressure between layers outside of the contraction zone, reducing their ability to transmit center torque, causing the cinching zone to expand to layers further from the core. 4. Cinching from internal gearing of roll: In center winding and unwinding, the roll's weight is supported on the core. The load of the roll's layers hanging over the core forms a high pressure area similar to a nip roller. As the roll rotates, the layers entering the high pressure zone will compress and elongate, rejecting some material like a nip roller. The excess material may fold over (like a nip roller accumulation wrinkle).
Solutions to MD Slip in Rolls	Solutions: 1. Increase the winding roll's torque capacity: (All these solutions will help both winding and unwinding cinching. <i>Solutions specific to help unwind-cinching after storage are in italics</i>) ▪ Increase tension, especially near the core (consider 'hard start') ▪ Increase friction coefficient ▪ Increase area and reduce the mechanical disadvantage of layers near the core with larger core diameter ▪ Increase crossweb thickness variations (including induced thickness variations from printing, coating, or knurling) ▪ Minimize core compression or shrinkage, including manage <i>paper core moisture content to ensure cores are dry</i>, use a core with sufficient stiffness for the internal roll pressures of your product (consider shifting to stiffer core) ▪ <i>Minimize shrinkage of layers, including substrates, coatings (including lateral flow of viscoelastic coatings, like adhesives), roughness or texture, and entrained air.</i> ▪ Minimize soft layers near the core from wrinkling or misaligned core starts. If surface winding, add center torque assist.

2. Decrease the applied center tensioning torque by reducing final diameter or final tension / more taper tension (both for center driven winders) or switch to surface winding.
3. Decrease the applied inertial torque by reducing final diameter or accel/decel rates.
4. Stop even the smallest amount of cinching to prevent cinching-induced cinching.

3C1: Web Shifts Laterally

Definition	TD shifting is a change in the web's lateral position as it runs through a process or on a roller. TD shifting should be divided into three categories. The first is watching the web's position on a roller throughout the contacting wrap angle. The second two are based on viewing the web at a single point monitoring its change in position. 1. The web runs helically around a roller, leaving a roller in a lateral position different from where it entered. 2. In running, the web makes a consistent offset shift laterally in the span between rollers. 3. In running, the web oscillates or wanders laterally in the span between rollers. In each of these cases, the web does not have to slip on any rollers. In the first case, the web does not slip relative to the roller, but travels around the roller like a stripe on a barber's pole.	
How to Measure	➤ Draw a line on the roller where the web edge runs and note if the web moves relative to the line. ➤ Use an edge sensor or line sensor from an automatic web guide to monitor the web's lateral position. ➤ Monitor an automatic web guide actuator motion to determine the shifting of the web entering guide.	
Problems	✓ Loss of lateral registration at slitting, coating, printing, laminating, and winding.	
Mechanisms and Root Causes plus Solutions for Helical Wrapping	Mechanisms and Root Causes	Solutions
	Roller Misalignment – Input web is misaligned relative to the downstream roller's direction of rotation (a.k.a. surface velocity vector).	Roller Misalignment – Align roller to each other, usually using machine centerline as a reference.
	Web Misalignment – The downstream roller in a web span is misaligned relative to the incoming web's entry angle.	Web Misalignment – Align the web to the rollers.
Mechanisms and Root Causes plus Solutions for TD Offset or Oscillating Shifting	Width Change – The web contracts and expands widthwise from tension increasing and decreasing, respectively.	Width Change – Reduce tension variations or cause of width change.
	Mechanisms and Root Causes	Solutions
	Initial Position TD Shift Causes	Initial Position TD Shift Solutions
	Shifted Layers in Unwinding Roll – Unwinding roll has shafted layers	Shifted Layers in Unwinding Roll – Ensure unwinding rolls have good

	without unwinding automatic guiding. <i>Unwinding roll has oscillated layers and no unwinding guiding.</i>	sidewall alignment and are correctly positioned on the unwinding core, shafts, and chucks.
	Threading Error – Web walks from initial threading position to running position.	Threading Error – Ensure the web is threaded correctly as close as possible to the web’s running position. Avoid unthreading equipment by leaving the process threaded, using a leader or trailer web when the product web cannot be left in the web line.
	Transport TD Shift Causes	Transport TD Shift Solutions
	Misalignment – Misalignment of roller, air turn bar, unwinding roll, or winding roll, especially on moving or eccentric elements such as dancer rollers, turret winders, out-of-roll or non-concentric unwinding or winding rolls. <i>Misaligned roller combined with web-to-roller traction variations causing the web to transitions from stick to slip behavior over time (shifting during stick time, slipping during slip time), especially from transitions due to inertia during acceleration or deceleration and from air lubrication as speeds increase.</i>	Misalignment – Ensure good alignment of all elements in contact with the web, including rollers, air turn bars, the unwinding roll, or the winding roll, especially moving or eccentric elements such as dancer rollers, turret winders, out-of-roll or non-concentric unwinding or winding rolls. Avoid transitions from stick to slip or good to poor traction by ensuring the Traction Safety Factor on all rollers is greater than one for all conditions including acceleration, deceleration, and high speed-to-tension cases. See Section 3A on solutions to avoid idler and driven roller MD slippage.
	Diameter Variations – diameter variations in roller, web on roller, or winding roll. <i>Crossweb diameter variations changing over time, such as a winding roll.</i>	Diameter Variations – Minimize diameter variations in rollers and winding rolls (especially in free span winding), including diameter variation of a web running with a coating or laminate between it and a roller acting like a diameter variation.
	Asymmetrical Web – Cambered or baggy web, especially in long spans and in air floatation handling. <i>Changing cambered or bagginess in a web, especially in long spans and in air floatation handling, including misaligned splices.</i>	Asymmetrical Web – Minimize web camber and bagginess. Ensure well-aligned splices.
	Uneven Nipping – Crossweb nip variations, especially with tangent	Uneven Nipping – Ensure crossweb uniformity in nipping. Wrap nips with

	entry into nips. <i>Changing crossweb nip variations, especially with tangent entry into nips.</i>	at least 5 to 10 degrees before the nip tangent point (more is better in this case) to create web-roller friction between nip variations in the entering span.
	Uneven Air – Crossweb variations in supply velocity of air nozzles or air turn bar, especially with long spans. <i>Unstable air supply in air nozzles or air turn bar.</i>	Uneven Air – Minimized crossweb variations in supply velocity of air nozzles or air turn bar, especially with long spans.
	Guiding TD Shift Causes	
	<i>Poor Tuning – Automatic guide PID loop is poorly tuned.</i>	Poor Tuning – Ensure web guide actuation rate and system gain is as high as possible and any actuation slop or measurement dead band are minimized.
	Excessive Input Errors – <i>Web walks on downstream roller from upstream source of lateral oscillation.</i> With automatic guiding, unwinding roll or input web has lateral step change (or oscillation) exceeding automatic guide's range or rate limits.	Excessive Input Errors – Minimize lateral errors starting at the unwind and working through the process to ensure good alignment downstream. Avoid exceeding automatic guides range. Center the web errors within the automatic guides range. If an unwinding roll is known to have shifting that will exceed an automatic guide actuation rate at normal process speeds (such as handling related telescoping), reduce the line speed when the unwinding roll gets to the high rate error layers to keep the error rate with the guide's actuation rate.
	Guide Geometry – Web guide geometry is non-ideal. <i>Automatic steering guide has under- or over-steering geometry, especially with unstable entry to pre-entry span ratio.</i>	Guide Geometry – Follow the recommended geometry for displacement and steering guides. Ensure displacement guides entry and exit spans are perpendicular to the guide's pivot plane. Ensure steering guides have long entry spans relative to web width, an exit span perpendicular to the roller pivot plane, and a pre-entry span shorter than the entry span.
	<i>Sensing Delay Time – Web guide</i>	Sensing Delay Time – Ensure the web

	<i>sensor is too far downstream.</i>	guide sensor is as close as possible to the web guide exit.
	Conflicting Guides – Upstream web guide has a different lateral guide point than downstream web guide.	Conflicting Guides – Ensure all automatic web guide in a process are set to the same lateral position relative to machine centerline.

3C2: Web Slips Laterally on Roller

Definition	TD Slippage is the lateral motion of any single point of the web relative to its initial point of contact on the roller during the angle of wrap. If MD slippage occurs (see Section 3A), TD slippage is also likely. TD slippage can be defined to occur when the web fails to follow the parallel entry rule.
How to Measure	➤ Measuring the difference between TD slippage and TD shifting can be difficult. ➤ A debris particle or burr on a roller will gouge the web during shifting, but create a scratch larger than the particle or burr in slippage.
Problems	✓ Scratching, abrasion (of web or roller) ✓ Unpredictable tracking ✓ Unpredictable web guiding ✓ Failure to spread wrinkle-prone web or slit strands Note: One benefit of webs slipping laterally on a roller – No wrinkles due to lack of friction.
Mechanisms And Root Causes	Root causes of TD slippage include: 1. Micro-slippage occurs to a degree on any roller if the web's strain change exceeds the web's ability to deflect the roller's surface. The most significant micro-slippage, both MD and TD, will occur on driven rollers that separate large tension changes. 2. Webs will laterally slip on a roller if the web-roller traction is insufficient to enforce the parallel entry rule, especially at steering roller with short entry spans and sometime displacement guides. 3. A web is highly likely to slip laterally on any roller with a speed mismatch relative to the web, whether oversped, undersped, or stopped. 4. Overly aggressive spreading of bowed roller or flat expander spreaders.
Solutions To Micro-Slippage Tension Change	First, the same changes that reduce MD slip from tension change will reduce TD micro-slip. 1. Reduce tension changes between zones. 2. Change tension set points, torques, or draws to create even tension into and out of drive rollers. 3. Reduce tension losses that create tension changes within tension zones, leading to confusion on what actual driven roller tension change is. 4. Place feedback devices near driven rollers to minimize confusion (at least on one side) of what the input or output tensions are. 5. When large tension changes are required, use multiple driven rollers to reduce the tension change over any one driven roller. Other solutions include: 1. Use a roller surface that can move with the web, such as a flex spreader roller or cloth wrap.
Solutions To Insufficient Traction	1. Increase web-to-roller traction (see the previous section for the many causes

To Enforce Parallel Entry Rule.	and solutions to MD slip). 2. Increase span length to reduce traction required to bend the web. 3. Reduce lateral shifting required by web guide by either matching the exit guide point to the entering web's position or shifting the web upstream to reduce the downstream guiding offset (the latter can be done proactively with coordinated guides). 4. Replace short span steering guides with displacement or sidelay guides where possible.
Solutions To Mismatched Speeds	1. Ensure driven rollers are accurate to within the stretch-ability of the web. 2. Avoid MD slippage (see previous section).
Solutions To Excessive Spreading Roller	1. Avoid setting excessive spreading percents, especially for high elastic modulus webs. 2. Avoid excessive wrap angles on bowed and flex expander rollers.

3D: Shift or Misalignment of Layers in a Wound or Winding Roll

Definition	Layers within a wound or winding roll slide laterally relative to each other or the core.
How to Measure	➤ For gross sidewall errors, simply measure the variation in position of layers to a straight edge. ➤ For small sidewall errors, use a dial indicator or profilometer to measure layer position relative to a straight reference. ➤ Unwind a roll without web guiding and measure the edge position with an edge sensor. ➤ With an automatically guided unwind, monitor the guide's actuator position as filtered indicator of unwinding roll lateral errors.
Problems	✓ Scratching, abrasion ✓ Roll sidewall alignment is out of specification ✓ Laterally shifted layers are easily damage in roll handling
Mechanisms And Root Causes	1. Web shifts prior to contact with winding roll. (See causes and solutions in Section 3A, 3B, and 3D.) 2. Layers shift within the winding roll ▪ air lubrication telescoping ▪ cinching-related telescoping ▪ deflection related telescoping (usually only slit rolls) 3. Core shifts relative to roll during winding. 4. PSA telescoping 5. Layers shift in roll handling or storage. 6. Layers shift within the roll during unwinding.
Solutions To Layers Shift Within The Winding Roll: Air Lubrication Telescoping	The outer layers of a winding roll will be fully lubricated when the entrained air exceeds combined roughness of the two web surfaces coming together at winding. Air lubrication at winding is a common problem of non-porous webs (paper, films, foils, poly-coated paper). Similar to air lubrication on a roller, a winding roll is fully lubricated when the entrained air thickness exceeds the web-to-roller combined roughness. Many products are prone to air lubrication problems if not managed properly. In winding, unlike on rollers, you rarely have the option to increase roughness or add air channeling grooves to eliminate the problem. Entrained air layer will also tend to be thicker at winding, especially since diameters may be many times higher than roller diameter and tensions may have to be set low, especially with taper tension to avoid tight roll or high torque defects. To make winding roll lubrication more complicated, the lubricated surface isn't a few inches of web-to-roller contact, but potentially hundreds of feet as the layers in a wound roll don't move from lubricated to good traction until either enough air bleeds out the sides of the roll or the pressure increase of multiple layers compresses the trapped air layer.

	The solutions to prevent winding lubrication includes many of the same options to prevent web-to-roller lubrication. Solutions: 1. Reduce the air layer thickness by limiting roll diameter, reducing speed, and keeping tension high. 2. Winding in a vacuum chamber eliminates the problem, but usually raises other problems as the benefits of air on winding friction and roll cylindricity are lost. 3. If possible in product design, consider a pattern or texture on one side of the web to manage air. 4. The most common air management solution is to winding with a roller nipping against the outside of the roll. The winding nip, found in all surface winding and many center winders, has two air management benefits. First, the nip load rejects, but does not eliminate, the amount of air that enters the winding roll. Second, under the footprint of the nip, the entrained air is compressed, creating at least a once per revolution traction point, enough to greatly reduce lateral shifting (a.k.a. air lubrication telescoping). 5. Physically restrain the web from shifting laterally with flanges, spool walls, or rotating elements. (This is certainly a bandage for the problem, but bandages are quite helpful.)
Solutions To Layers Shift Within The Winding Roll: Cinching-Related Telescoping	As the name implies, cinching drives this defect, where the torque applied to the roll exceeds the roll's torque capacity. A roll's torque transmission capacity is calculated throughout a roll by multiplying any layer's internal radial pressure by its area of contact and side A-to-B coefficient of friction. In winding roll, the primarily applied torque is from center winding drive creating tension at the roll's outer diameter; however, inertial torque during acceleration or deceleration should also be considered. The onset of cinching related telescoping is most commonly seen as a roll grows in diameter and first appears in the layers near the core and the outer wraps. The problem occurs as center winding rolls growing diameter require more torque to create tension. The cinching onset begins near the core since these layers have less area and are at a mechanical disadvantage to the outside of the roller (creating a radius-square effect, with outer layers having more torque capacity). Since the relationship of tension to pressure is non-linear, a small increase in tension can greatly increase torque capacity. Also, a roll with flat thickness profile will have less torque capacity than a roll with gauge variations. Thick, low friction webs are the most prone to cinching-related telescoping. Solutions: 1. Increase the winding roll's torque capacity: increase tension, especially near the core, increase friction coefficient, reduce roll buildup with either larger cores or smaller final diameter, and increase crossweb thickness variations (including induced thickness variations from printing, coating, or knurling). 2. Decrease the applied torque by reducing final diameter, final tension, and

	accel/decel rates. 3. Physically restrain the web from shifting laterally with flanges, spool walls, or rotating elements. (This is certainly a bandage for the problem, but bandages are quite helpful.)
Solutions To Layers Shift Within The Winding Roll: Deflection Of Slit Roll Winding Shafts	Much of winding is based on 3-in inner diameter cores; however, as the load (the vector sum of roll weight, nip load, and web tension) on the shaft increases, so too will the shaft deflection. This may not cause a problem in the center of a symmetrically deflection shaft since this portion of the shaft will still be parallel to the other rollers. However, the angles at the ends of the shaft will create an angle that will cause the web to walk laterally or create enough side load to cause the winding roll to collapse like a leaning tower. Short, small diameter rolls will usually be no problem since they don't create high loads on the shaft. Wide rolls are also often no problem since the tensioned layers add to the bending stiffness of the shaft. The greatest shaft deflection problem occurs with winding several narrow, larger diameter rolls, especially of high density webs, such as uncoated films and foils. Solutions: 1. Use a center cam arm to support the shaft near its center. 2. Nip the slitting shaft from the bottom to offset the gravity load. 3. Avoid excessive nip loads. 4. Extend the shafts and provide additional rigid supports or counter bending from the shaft ends. 5. Cut the input jumbo in half and slit half the number of slit rolls per cut. 6. Use a core-dependent differential shaft since they usually have higher bending stiffness than core-lock differential shafts.
Solutions To Core Shifts Laterally Cases	1. Chuck support: The core is relatively unlikely to slip laterally, since most chucks have collars that butt up against the core's side. However, in central drum and surface winders where the core moves away from the fixed roller as the roll builds, if the support arms are misaligned, they will shift the core laterally changing the lateral registration of the core to the outside of the winding roll. 2. Single-roll shaft support: For wide webs winding, the most common is operator failure to inflate shaft manual pressurizing systems. When properly pressurized or engaged, cores slip in the machine direction when the applied torque exceeds the frictional torque limit between the core and shaft. Machine direction slippage is almost always accompanied by laterally slippage. Core slip is a common problem for plastic and metal cores running on shafts designed to bite into paper cores (use of rubber elements can help). Core slip is more likely at higher applied torque from increasing tension or roll diameter. 3. Differential winding shafts: Differential shafts must slide in the machine direction relative to the winding shaft to properly function, but the challenge is how to hold the core laterally while allowing it to slip in the machine direction. Side-loading differential shafts attempt to solve this problem by

	physically stacking cores and spacers to hold the cores laterally. Radially-loading differential shafts usually don't include stacks of cores and spacers, so they need some positive mechanism to hold the core laterally and how a specific differential shaft accomplishes this is often one of the key features in their patent-able design. Lateral restraints include pins and bearing to physically contact the core flanks, ball bearings that indent groove the inside of a paper core, and several that have laterally stacked core-grabbing elements. 4. Shaftless, chuckless systems: In two-drum surface slitter-rewinders, some systems have no physical restraint to hold the core position. As the roll builds, the core is free to move laterally if the roll shape is uneven. The two dominate core shifting mechanisms are: 1) a roll building into a cone-shape due to asymmetrical crossweb thickness, 2) width recovery of low modulus webs expanding laterally as winding pressure drive layers within the roll towards the core, dropping their circumferential tension and recovering their untensioned width. In a single roll, this would form a pear shaped roll profile, but in tightly spaced slit rolls with unconstrained cores, the width recovery pushes the cores outward, progressively worse from the center to the edge cuts.
Solutions To Layers Shift In Roll Handling Or Storage	Cause: Opposing forces applied to the roll exceeds friction with roll. Viscoelastic adhesive layers creep laterally in response to asymmetrical high pressures within a roll. Solutions: 1. Increase internal friction by increasing internal roll pressures, area, or web side-to-side coefficient of friction. Internal roll pressures increase with stiffer cores, higher winding tension, less tension taper, higher nip load, lower speed (less entrained air), lower product tensile modulus, and stiffer stack modulus. 2. Reduce applied loads, including friction to remove roll from shaft or chuck, inertial loads from quick accel/decel motions, and gravitational loads from handling rolls vertically. 3. Apply loads where rolls have more friction (e.g. push or pull a roll from the core or layers near the core). 4. Avoid lateral loads (e.g. unload a roll in the machine direction, rather than laterally). 5. For adhesive telescoping: Reduce roll pressure by methods listed in #1 above. Reduce viscosity and thickness of adhesive material properties. Reduce viscosity by storing rolls at reduce temperatures. Oppose lateral creep with loaded side shield or packaging. Reduce lateral motion with thickness symmetry.
Solutions To Layers Shift Within The Roll During Unwinding	Cinching usually drives this defect, where the torque applied to the roll exceeds the roll's torque capacity. The first rule of unwinding is to unwind at a tension lower than the roll was wound at, typically the final winding tension. The true goal of this rule is to avoid applying a torque at unwinding higher than was applied at winding to avoid cinching and cinching-related telescoping.

Since the relationship of tension to pressure is non-linear, a roll with flat thickness profile will have less torque capacity than a roll with gauge variations. Keeping unwinding torque below a roll's winding torque does not guarantee you won't exceed an unwinding roll's torque capacity and induce cinching-related telescoping since a roll's torque capacity may change during extended storage. Torque transmission capacity is directly proportional to the internal pressure in a roll.

In storage, the escape of wound in air, the viscoelastic flow of webs or coatings, the inelastic yielding of surface roughness, and many other factors may drop the pressure within a stored roll over time, dropping its torque capacity and increasing the likelihood of cinching and telescoping at unwinding.

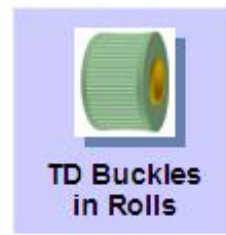
Thick, low friction webs are the most prone to unwind telescoping.

Besides cinching-related telescoping, lateral shifting may be induced by hyperactive sidelay web guiding or uneven stresses associated with a deformed slit edge or asymmetrical thickness profile.

Solutions:

1. Increase an unwinding roll's torque capacity at winding: high tension, especially near the core, high friction coefficient, less roll buildup by either larger cores or smaller final diameter, and increase crossweb thickness variations (including induced thickness variations from printing, coating, or knurling).
2. Decrease loss of torque capacity in storage by reducing air entrainment, changing material properties or storage conditions (such as refrigeration) to reduce viscoelastic flow or shrinkage, and reduce storage time.
3. Reduce lateral or asymmetrical forces on the roll from sidelay guiding, slit edge quality, and gauge profile.
4. Physically restrain the web from shifting laterally with flanges, spool walls, or rotating elements. (This is certainly a bandage for the problem, but bandages are quite helpful.)

IV. Solutions to Buckled Webs Problems



4A: The Web Buckles Between Rollers (Troughing)

Definition	The web is out of plane in the span between rollers.
How to Measure	➤ Visual inspection viewing web with low angled lighting. ➤ Noting frequency and amplitude of troughs can help separate temporary from permanent troughs. ➤ A low angle line scan laser can highlight troughing severity.
Problems	✓ Amplitude of buckling is enough to contact nozzles or slots. ✓ Amplitude of buckling is out of the field of view of camera or laser inspection systems. ✓ Buckled shape is locked in if present during curing, drying, or quenching process.
Mechanisms and Root Causes	1. Web is bending left or right (see mechanisms of shear wrinkles below). 2. Two sides of the web are tracking towards each other (see mechanisms of tracking wrinkles below).
Solutions to buckling in spans	1. Increase Web Stiffness – Use a thicker or high modulus web. Avoid modulus lowering effects such as moisture in paper webs and heat in thermoplastic film webs. 2. Shorter Span Lengths – For a non-wrinkling condition (defined as the web is flat while in contact with rollers), the troughing will flatten out in the portion of the span close to the roller entrance and exit contact points. If two rollers are close enough together, the non-troughing zone may extend the entire span. 3. Stop Bending Cause – Stop the cause of bending or tracking to centerline listed below under shear and tracking wrinkles. 4. Use Shape Stiffening – Buckling at nozzles: Induce some shape into the web with alternating air nozzles or roller near the nozzles to create shape stiffening. 5. Place Rollers in Flat Web Critical Position – For buckling critical positions, such as slots and inspection, place a roller near a slot's entrance and/or exit or as close as possible to the point of inspection to prevent out-of-plane shape or motion with shape stiffening and roller support. 6. Use Spreading Mechanisms – A spreader roller, properly installed, can create a greater trough-free zone than an ordinary idler, but will have limited

benefits a short distance upstream of the spreader and almost immediately downstream. A clip or pin tenter frame can grab and maintain the web width or diverge and spread the web.

Buckled Web on Rollers

This section covers buckled web defects that occur while the web is near or on a roller.

4B1: Accumulation Wrinkles

Definition	All or a portion of the width of the web is under machine direction compression and buckling as it passes over a roller, through a nip point, or enters a winding roll.
How to Measure	➤ Most wrinkles are measured by viewing or counting the creases in the web. ➤ For all wrinkling defects (buckling on rollers), observe and note the angle and position any troughs in the span immediately upstream of where the wrinkles form, the angle and crossweb position of the wrinkles, and whether the wrinkle shifts or walks laterally.
Problems	✓ Creases ✓ Web breaks ✓ Coating and laminating defects (including uncoated areas, bubbles, and delamination) ✓ Winding defects
Mechanisms and Root Causes	All causes of machine direction buckling or wrinkling fall under the wrinkle classification of ‘accumulation wrinkles’ where the entire width or a portion of the web’s width is under machine direction compression beyond it’s buckling criteria (usually low due to lack of shape stiffening in the machine direction). Accumulation wrinkles most often form immediately upstream of any system that meters the web feed rate (speed), including non-slipping idler roller, nipped rollers, and winding rolls (and can also include tenters or belts). These web metering systems have a strong tendency to feed a set rate, in surface length per time usually defined in feet or meters per minute or inches per second. If the amount of web delivered to the metering system exceeds the rate the web is metered through the system, the differential will drive the web towards compression. A negative draw, where the input speed is higher than the output speed does not automatically lead to compressive stresses. If the input web is tensioned, small amounts of negative draw will lead to lower tension. If the negative draw is greater than the input web’s strain then the web will go into compression and buckling. Baggy webs are especially susceptible to accumulation wrinkles since, by definition, some portion of the web’s width is running at below average or zero tension and is easily pushed into compression and buckling. As with wrinkling conditions, the compression stresses required to buckle the web must be supported by forces (usually frictional forces, but gravity and inertia forces may be enough) from both opposing directions (upstream and downstream in this case). The top causes of accumulation wrinkles are: 1. Baggy webs and nipped rollers 2. Overfed nips in calendar stacks, nipped rollers with large pre-nip wraps, and

	surface winders 3. Delamination of laminates ahead of rollers (especially small diameter rollers) 4. Laminate air bubble tunneling / accumulation 5. Air bubbles in nipped winding 6. Nip-induced compression at unwinding
Solutions To: Baggy Webs and Nipped Rollers	1. Eliminate nips wherever possible (use high friction, large wrap angle roller or vacuum roller) 2. Increase tension to pull out bagginess. 3. Use an aggressive spreader roller (usually a bowed roller) to pull the web laterally and through the Poisson's effect, contract the long lanes of bagginess. 4. For one-sided bagginess, use an adjustable roller to create a longer web path on the loose side to tighten the web immediately before the nip. 5. Reduce the nip load to allow the web to slide through. 6. Lubricate the nip (if coating or printing, accumulation wrinkles may stop once coating or printing begins).
Solutions To:Overfed Nips	1. Reduce the traction between the web and roller ahead of the nip point by reducing wrap angle, tension, coefficient of friction, or lubricating with air or liquid. 2. Avoid nipping the web twice on the same roller. 3. If nipping twice on one roller is required, such as in a calendar stack, use 'fly' rollers to peel the web off the calendar roller after the nip point and minimize the wrap angle before the next nip.
Solutions To: Delamination At Rollers	1. Use large diameter rollers 2. Delay contact with rollers immediately after laminating to allow green strength bond to increase 3. Bend with most flexible web away from the roller or outside in the winding roll. 4. Use a laterally ridged roller to burp through small buckles to avoid product damaging larger buckle.
Solutions To Laminate Air Bubble Tunneling / Accumulation	1. Make one web porous or perforated. 2. Lance the bubble at start up or as it grows too large. 3. Eliminate the source of the air bubbles (e.g. high nip pressure). 4. Increase bond strength to prevent bubble tunneling delamination and accumulation. 5. Curve around rollers with the lower tensioned laminate layer out. 6. Use a laterally ridged roller to burp through small buckles to avoid product damaging larger buckle.
Solutions To Air Bubbles In Nipped Winding	1. Reduce the entrained air with high nip loads, lower speeds. 2. Make the web porous or perforated to allow the air through the web. 3. Wind with an interleave web that can absorb the air layer.

	4. Make the product rough or textured to allow the air to channel away. 5. Use a 'vent-a-groove' roller.
Solutions To Nip-Induced Compression At Unwinding	1. Avoid nipping unwinding rolls 2. Minimize nip load at unwinding. 3. Use a laterally ridged roller. 4. Use a belt instead of a roller
Comments on Spreader Rollers And Accumulation Wrinkles	Since accumulation wrinkles are driven by machine direction compression and spreader rollers are designed to create transverse direction tensioning, it does not seem that a spreader roller would be an appropriate tool to prevent accumulation wrinkles. However, aggressive spreader rollers are effective in some cases, especially if the accumulation wrinkles are not full width. Through the Poisson's effect, the lateral tensioning of the web will create some machine direction contraction, most likely in the low or zero tension baggy lanes and help reduce accumulation wrinkles associated with bagginess.

4B2: Shear Wrinkles

Definition	The web is buckled in the crossweb (transverse) direction forming creases running in the machine direction (MD) or slightly angled from MD on a roller or as it enters a winding roll (wrinkling) due to lateral bending in the upstream span.
How to Measure	➤ Most wrinkles are measured by viewing or counting the creases in the web. ➤ View wrinkling in high speed processes by projecting a visible laser beam crossweb at a small spacing above the roller's surface. Any wrinkling that buckling into the laser's path will reflect the laser light for easy observation. ➤ For all wrinkling defects (buckling on rollers), observe and note the angle and position any troughs in the span immediately upstream of where the wrinkles forms, the angle and crossweb position of the wrinkles, and whether the wrinkle shifts or walks laterally.
Problems	✓ Creases ✓ Web breaks ✓ Coating and laminating defects (including uncoated areas, bubbles, and delamination) ✓ Winding defects
Mechanisms and Root Causes	All shear wrinkles have three pre-requisites. A. A mechanism that bends the web laterally. B. Sufficient friction to bend the web laterally. C. Sufficient friction to hold the web in a buckled form while on the roller. The top mechanisms of bending that lead to shear wrinkles are: 1. Roller misalignment 2. Tapered roller diameter with good traction or poor traction 3. Uneven tensioning from left-right nip load variations 4. Short term lateral shifts in the web, either step or sinusoidal shifting 5. Uneven tension from differential peeling off upstream roller or unwinding roll. 6. External forces pushing laterally on the web.
Solutions to Roller Misalignment	1. Ensure foundation will support equipment weight and is stable over time. 2. Ensure equipment framework is rigid and stable. 3. Symmetrical design can prevent thermal expansion misalignments. 4. Design equipment and rollers systems with considerations for ease of alignment, re-alignment, and holding alignment. 5. The first step to aligning pivoting rollers or shafts is to align the pivot axis. Make the pivot shaft or axis easy to align. For rollers and shafts that are intended to hold alignment while pivoting (e.g. dancer rollers, turret winders), use keyways to ensure arms are parallel. 6. For dual-end supported rollers, shafts, or pivot shafts, use split cap roller mounts to ease removal and replacement into aligned position. 7. For cantilevered rollers mounted on a reference plate, use large base flanges

	to allow horizontal and vertical alignment. Avoid three-bolt tripod adjustment. - For cantilevered rollers without reference plate, use either 1) low-deflection cantilevered beam and a dual-end supported roller or 2) a one-side dual-supported shaft. - Measure level and tram with precision alignment equipment capable of measuring less than 2 mils per foot errors. This includes optical transits with rulers and mirrors, laser alignment, pi tapes, and tramping sticks. This does NOT include tape measures or carpenter's levels. - Drill and pin aligned rollers. Don't trust bolt friction to hold a roller in place.
Solutions to Tapered Rollers and Rolls	- Ensure rollers are cylindrical to less than one half of the web strain from tension. For polyester and paper products running at 0.1%, the roller should be cylindrical with error less than one mil per inch of radius (i.e. a 4-in idler should be +/- 1 mil, a 12-in roller +/- 2 mils). - Measure roller diameter variations with a pi tape to ensure cylindricity. - Check rubber roller wear in your preventive maintenance plans. - For rubber rollers, take nip impressions of the rubber roller sitting on an optically flat plate. - Mark all rollers that have intentional diameter variations. - Check winding rolls for diameter variations causing shear wrinkles. Reduce winding roll diameter problems with improved product thickness profile or allowing more air to winding into the roll.
Solutions to Uneven Nipping	Check nip footprint or pressure Uneven footprint causes: - Force load is uneven left to right. - Either roller has left-right diameter variations - Nip pivoting mechanism is misaligned and does not have freedom align to fixed roller. - Product has left-right thickness variations. - Compliant roller has hardness variations. - Nip system is blocked on one side by uneven roller stops.
Solutions to Short Term Lateral Shifts	Any lateral error can be characterized by magnitude and rate of shift. Wrinkles from shifting are most likely when the shape of the web between rollers is similar to the shape that causes wrinkling from misalignment and tapered rollers. Example A: If a web-roller combination is sensitive to 10 mils per foot (width) misalignment in a 2-foot span, then a lateral shift of 10 mils per foot (length) that lasts for two feet will likely wrinkle in the same span. Luckily, most web shifting occurs over much longer web lengths Example B: If for the same web-roller combination, a web makes a step change in position, shifting 0.2-in in one second at 500 fpm (100 ips), that is 200 mils over 100 ft, or 2 mils per foot. The web will likely not wrinkle since this is five times lower than what would cause a wrinkle in the previous example. The web is also fairly insensitive to high frequency oscillations. If the web shifts 10 mils every 5 inches over a 2-ft span, the rate of error is high (24 mils per foot),

	but the net bending a 2-ft span may be zero since the web moves left and right several times within the span. 1. Eliminate the causes of lateral shifts with magnitude and wavelength that are prone to causing wrinkles. 2. Slow down processes or cut out when unwinding layers in a roll with sharp lateral shifts. 3. Reduce traction on rollers.
Solutions to External Air Forces	Air flow, both ambient and within force-air dryer ovens, is one of the top external forces that may shift a web laterally. Prevent air flow induced lateral shifting by: 1. Avoid long unsupported spans. 2. Ensure traction on rollers. 3. Design oven air returns to create symmetrical flow with preferred design having the air returns in the center of the web width. 4. Avoid excessive amplitude in sinusoidal air floatation ovens. 5. Avoid running one-side baggy web in floatation ovens. 6. Use steering style web guide as the first roller at the exit of ovens, but ensure the span in to the steering guide is long (entry length > 3 web widths).
Solutions by Reduce Traction	Since shear wrinkles require web-roller traction in two of three prerequisites, reducing traction can often eliminate shear wrinkles. Four methods prevent wrinkles by traction control. (These were covered above in the Eliminating Wrinkles notes.) 1. Force the roller into slip 2. Low traction with existing web path 3. Lower traction with a new web path 4. Add a roller to reduce wrap angles.
Solutions by Changes in Span Lengths	As mentioned below in the Eliminating Wrinkles Action Plan notes. Adjust span lengths to eliminate wrinkles. For shear wrinkles, take out roller or move their location to create longer spans. For tracking wrinkles, add rollers to shorten the web spans. In most wrinkles caused by misalignment, the wrinkle forms either on the misaligned roller or on the roller immediately of the misaligned roller. There is one unique case where a misaligned roller can create a wrinkle on the roller immediately upstream. This scenario is referred to as either an unstable web path, especially regarding steering-type web guides, or negative steering. Unstable path or negative steering is eliminated by have a long span upstream (a.k.a. the entry span) of the misaligned rollers, at least long relative to the span ahead of the upstream roller (a.k.a. the pre-entry span).
Solutions by Spreaders and Anti-Wrinkle Rollers	Shear wrinkles are effectively prevented by all types of spreader rollers, including bowed, flat expanders, flex expanders, concave, D-bars, and hyper-crowns.

4B3: Tracking/Gathering (MD) Wrinkles

Definition	The web is buckled from compressive stresses in the machine direction forming creases in the crossweb (transverse) direction or slightly angled from the TD on a roller or as it enters a winding roll (wrinkling) due to a tracking mechanism where two sides of the web track towards each other.
How to Measure	➤ Most wrinkles are measured by viewing or counting the creases in the web. ➤ View wrinkling in high speed processes by projecting a visible laser beam crossweb at a small spacing above the roller's surface. Any wrinkling that is buckling into the laser's path will reflect the laser light for easy observation. ➤ For all wrinkling defects (buckling on rollers), observe and note the angle and position any troughs in the span immediately upstream of where the wrinkles form, the angle and crossweb position of the wrinkles, and whether the wrinkle shifts or walks laterally.
Problems	✓ Creases, web breaks ✓ Coating and laminating defects (including uncoated areas, bubbles, and delamination) ✓ Winding defects
Mechanism:	All tracking wrinkles have three pre-requisites. A. A mechanism that bends the two sides of the web towards each other. B. Sufficient friction to bend the two sides of the web laterally. C. Sufficient friction to hold the web in a buckled form while on the roller.
Root Causes, and Solutions	The top mechanisms of tracking that lead to wrinkles (and solutions to the wrinkles) are: 1. A convex roller profile with good traction The most common convex (crowned) rollers are at nip points to compensate for nip roller deflections. Avoid large wrap angles to prevent the convex shape from gathering the web. 2. A concave roller profile with poor traction Ensure concave rollers have sufficient traction, through good wrap angle, tension, coefficient of friction, and grooving or texture to prevent lubrication. 3. A bow or deflection in the direction of the incoming web Minimize deflection due to gravity, tension, nip loads, or incorrect adjustment of a bowed roller. Avoid deflection by using sufficiently stiff rollers for the application. Large diameter is the strongest change to make a roller stiffer. Changing from aluminum to steel will have a 3X reduction in deflection. Avoid wrapping nip roller 90-degrees prior to the nip point so the entering web path see less than 100 percent of the negative bow from deflection. 4. Web expanding in the span upstream of a roller Eliminate the source of width expansion or use a spreader roller at the end of spans where expansion occurs. 5. Baggy center web in long spans.

	Eliminate the bagginess or eliminate long spans. 6. Tension change from high to low. Reduce tension transients over time (see section 1B). 7. Excessive twisting in a span. Avoid excessive twisting or use a spreader at the end of the twisting span.

4B4: On-Roller Expansion Wrinkles

Definition	The web is buckled from compressive stresses in the crossweb (transverse) forming lines or creases in the machine direction on a roller due to width-wise web expansion while in contact with the roller (or by width-wise contract of the roller surface).
How to Measure	➤ Most wrinkles are measured by viewing or counting the creases in the web. ➤ View wrinkling in high speed processes by projecting a visible laser beam crossweb at a small spacing above the roller's surface. Any wrinkling that is buckling into the laser's path will reflect the laser light for easy observation. ➤ For all wrinkling defects (buckling on rollers), observe and note the angle and position any troughs in the span immediately upstream of where the wrinkles forms, the angle and crossweb position of the wrinkles, and whether the wrinkle shifts or walks laterally.
Problems	✓ Uneven heat transfer on heating or cooling rollers. ✓ Permanent creases, if nipped while buckled. ✓ Uneven coating, especially with radiant coating such as vacuum metalizing.
Mechanism	All tracking wrinkles have two pre-requisites. A. A mechanism expands the web width while in contact with a roller. B. Sufficient friction to hold the web in a buckled form while on the roller.
Root Causes and Their Solutions	The top mechanisms of on-roller expansion that lead to wrinkles are: 1. Tension change on a driven or clutched roller from high to low tension Reduce the tension change by changing the set point in one or both zones. Reduce the tension change due to roller and other tension losses (see causes under section above on tension variations within a zone). Make the tension change over a series of rollers with progressive decreasing draw. 2. Moisturizing paper (and other materials with high coefficient of hygroscopic expansion) Moisturize paper in spans, not on rollers. Avoid condensation on chill rollers by either raising chill water temperature (at least until hot web can do so) or decreasing the environmental humidity. 3. Heating films and foils (and other materials with high coefficient of thermal expansion) Laterally stretch extensible films prior to rollers where the film will be heated (heated rollers, backup rollers for vacuum metallization). Instead of a single heated roller, heat the film or foil over a series of rollers.

Wrinkle Elimination Action Plan

Find the Wrinkle Origin	1. Walk the process from unwind (or most upstream point) to rewind (or most downstream point). 2. Find the point where the web goes in wrinkle-free and comes out wrinkled. Try to identify the exact roller of the first wrinkle. Use a low angle light or pin-laser to assist in seeing the wrinkles and troughs. 3. Sketch an elevation diagram of the roller web path where the wrinkles occur. Sketch a top view (or view perpendicular to the web's plane) showing the direction of the troughing, the lateral position where the wrinkles are forming, if the wrinkles are moving laterally or stationary, and if moving, in what direction. 4. Note features of the rollers where the wrinkles occur, including a few rollers upstream and downstream. Note the material and condition of the roller surface, the way the roller mounts to the equipment framework, the rigidity (or lack thereof) of the equipment framework, and if the roller is part of a moving element (turret winder, accumulator, nipping roller). 5. Check the roller alignment of the rollers where the wrinkles occur, including a few rollers upstream and downstream. Level is easiest to measure. Tram is more difficult. Use a pi tape or tramming stick for quick tram measurements. Check roller-to-roller tram in the sequence of the web path. 6. Run the process at low speed. Some wrinkles will go away with the air lubrication of high speed handling, but a good goal is to be wrinkle free at all speeds.
Eliminating Wrinkles – Immediate Fixes	1. Change the input roll – Understand the contribution of input web quality to wrinkles. Measure bagginess or camber of rolls that do and do not wrinkle. If possible, flip the input roll end for end to see if the wrinkles flip as well. 2. Change the local process tension – Many wrinkles will go away at either higher or lower tension. Higher tension provides tension stiffening, increasing the compressive stress needed to buckle the web. Lower tension may reduce traction between the web and roller, reducing the force available to bend the web or hold the web in a buckled shape while on the roller. Make sure that changing the local tension doesn't upset the process elsewhere (e.g. Too low of tension may increase slip, wander, scratching, or loose wind defects. Too high of tension may increase likelihood of web breaks, cause slip at driven rollers, or tight wind defects). 3. Try tape collars at both web edges – Two bands of masking tape running under each web edge on a wrinkle-prone roller is a converter's tool that costs pennies and saves millions each year. It is a bandage to the problem, but everyone knows bandages are important in life. 4. Modify the friction coefficient of the wrinkle-prone roller – Changing the surface of a wrinkle-prone roller to a low friction or smooth, easy to lubricate surface can stop wrinkles with the same mechanism as lowering the tension. Low web-roller traction means less ability to pull the web laterally or track to the center and less force to hold the web in a buckled shape on the roller. 5. Stop the wrinkle-prone roller – Use the friction-circle effect to flatten the wrinkle. When a roller is turned into a drag bar (by forcing it into slip, whether stopping, undersped, or oversped), all the available traction is used in the machine direction leaving none for span bending or lateral forces needed to hold the web in a buckled shape on the roller. (This solution is not viable for scratch sensitive

	products.) 6. Try a new web path with existing roller – Change the span length or wrap angle of the wrinkle-prone roller by bypassing rollers or changing the order of threading.
Eliminate Wrinkles – Longer-Term Fixes	1. Understand and eliminate the wrinkle cause – Diagnose the wrinkle cause from the visual clues. Is the wrinkle a shear, tracking, expansion, or accumulation wrinkle (or some combination of these)? Check and correct any misalignment, diameter variations, web bagginess / camber, and nip variations. 2. Adjust span lengths – For shear wrinkles, take out roller or move their location to create longer spans. For tracking wrinkles, add rollers to shorten the web spans. 3. Adjust wrap angles – Sometimes adding a roller to replace a single 90-degree wrapped roller with two 45-degree wrapped rollers will eliminate a wrinkle. When adding a roller to share wrap angles of a roller, instead of splitting the existing wrap angle in half between two rollers, try giving the first roller only enough wrap to ensure it doesn't slip and the remaining to the second roller. A low wrapped idler in front of a wrinkle prone location is called a kiss roller. Kiss rollers usually only have 5-10 degrees of wrap. Also try adding a kiss roller immediately upstream of the wrinkle-prone nip or winding roll. 4. Try gap or pack rollers at winding – Many winding wrinkle problems are due to winder designs that intended for wrinkle-insensitive products (e.g. thick papers). For wrinkle-prone webs, never use free-span winding where the misaligned, diameter-varied core or winding roll controls a long span. Use or add gap and nip roller winding to present a wrinkle-free web to the winding roll. 5. Try an adjustable roller to tighten the web left or right – An adjustable roller immediately upstream of a nip point can tighten a web with one loose edge. Ensure the adjustable roller is marked to show where it well-aligned position is so it can be moved back during normal operation. 6. Spreaders and anti-wrinkle rollers – When all else fails, install a spreader or anti-wrinkle roller (or drag bar) in place of or immediately upstream of the wrinkle-prone roller.

Buckled Web in Wound Rolls

This section covers buckled web defects that occur after the web enters the winding roll.

4C1: The Wound Roll TD Buckles

Definition	Machine direction compressive stresses (a.k.a circumferential or hoop stresses) have exceeded the buckling criteria of the layers within a wound roll, forming transverse direction buckles or wrinkling in layers of a wound roll. TD buckles may be throughout a roll, but more commonly occur in a particular region (e.g. near the core, on one side, between gauge bands). TD buckling defects have many names, including core buckles, starring, spoking, corrugations, crossbuckles, softbands, gapping, and peaking. TD buckles occurring near the core radiate in small or large triangular patterns and are usually called starring or spoking defects. Starring is a paper industry term, where the buckled layers in rolls of thicker papers may form 2 to 5 large triangular shaped patterns that may radiate 2 to 20-inches from the core. Spoking is a film industry term, where the buckled layers in rolls of thinner films will form 10 to 50 narrow spike-like patterns that usually only radiate 0.25 to 1-inche from the core.
How To Measure	➤ TD buckles in rolls are measured by direct observation. ➤ Note width and radial positions where TD buckles occur – Full width vs. in specific lanes across the roll's width, throughout the roll or only in specific radial positions, such as in layers near the core or at the roll's outer diameter layers.
Problems	✓ TD buckles may become permanent deformations in the web and create functional or optical defects in final products, especially in laminated or coated products.
Mechanisms And Root Causes	1. Cross-buckles and softbands: TD buckles that form in the wound roll where lanes of below average thickness web or coating wind in the same lateral position for many layers. The layers in the thin lanes are unsupported and fall towards the core, relieving their hoop tension, and with enough radial change, go into hoop compression and buckling. 2. Starring and spoking: TD buckles occurring near the core radiate in small or large triangular patterns and are usually called starring or spoking defects. Starring/spoking form under the combination of a) a core with less stiffness than the layers near the core and b) high pressure compressing the core changing its radius by more than the strain in the web. Starring is a paper industry term, where the buckled layers in rolls of thicker papers may form 2 to 5 large triangular shaped patterns that may radiate 2 to 20-inches from the core. Spoking is a film industry term, where the buckled layers in rolls of thinner films will for 10 to 50 narrow spike-like patterns that usually only radiate 0.25 to 1-inche from the core. 3. Soft start buckles: Viewed from the side or upon unwinding a roll, buckled layers are found near, but not immediately at the core. Where starring and spoking are due to core compression, soft start buckles are caused when the

	initial layers of a winding roll have low tension, wrinkles, or misalignment buckles and are subsequently compressed by the pressure of the wound roll. 4. Softbands and crossbuckles: Buckles in low thickness bands (soft bands), across the full width of the roll, or in the most dramatic case, the entire roll sags into an eggs shape. 5. Peaking, gapping, and delamination (specific to pressure sensitive adhesive products): When adhesive layers shrink in the thickness direction by post-wind drying or lateral viscous flow, layers or the full roll may shift from hoop tension to hoop compression. In lined PSA products, this may create defects similar to starring, spoking, softbands, and crossbuckles, but it may also cause delamination within the roll. In self-wound PSA products, since the layers are bonded together with adhesion, extreme loss of tension will not lead to sagging or egg-shaped rolls, instead forming buckling throughout the roll. In narrow rolls, the in-roll delamination may traverse the entire width of the roll creating a defect called gapping or tunneling. If the high compressive hoop stresses are throughout the entire roll, the exterior of the roll may lose its roundness, forming buckled ridges (referred to as peaking or gearing). 6. Core collapse, core crushing: When the pressure on the core exceeds its buckling strength, the core buckles and the layers of the roll follow.
Solutions To Cross-Buckles and Softbands	1. Wind on larger cores or smaller buildup ratios. 2. Wind at high speed to allow air to be captured in the thin lanes 3. Reduce thickness variations 4. Wind tighter. 5. Reduce stack modulus web or wind with interleave with lower stack modulus. 6. Oscillate thickness variation in wound roll.
Solutions To Starring And Spoking	1. Wind on a core with high stiffness (resistance to compression from buildup of roll core pressure). 2. Reduce the pressure exerted on the near-core layers with taper tension, limiting roll buildup, decreasing stack modulus, or winding on larger cores.
Soft Start Buckles	1. Ensure the initial layers of the wound rolls are especially tight and free of wrinkles, misalignment, core scrap, or excessive entrained air.
Solutions To Crossbuckles And Softbands	1. Reduce crossweb thickness variations, especially thickness profiles that are asymmetrical (one side thick, one side thin) or create 'valleys' where two thick gauge bands have a band of average or below average thickness between them. 2. Reduce diameter variations created by thickness variations with lower stack modulus, either by product design, lower winding tensions, or more entrained air. 3. Manage entrained air in winding to minimize radial compression in a roll over time as air escapes. 4. Oscillate thickness variations with flat process shifting or blown film bubble rotation.

	Minimize material or process conditions that cause layers within a roll to lose thickness over time, including coating, laminate layers, and surface roughness or features.
Solutions To Peaking, Gapping, And Delamination (Specific To PSA Products)	- Reduce thickness change in adhesive layer through formulation change, reduce thickness, storing the rolls in a cool environment, or reducing the storage time. - Change the winding profile to start and end with higher tensions. - Increase core stiffness.
Solutions To Core Crush-Related Buckles	- Wind on core with high strength (different from high stiffness). - Reduce the pressure exerted on the core via lower starting tension, more taper tension, and stack modulus.

4C2: The Wound Roll MD Buckles.

Definition	Transverse direction compressive stresses have exceeded the buckling criteria of the layers within a wound roll, forming MD lines, buckles, or wrinkles in the layers of a wound roll (often called MD lines or tin-canning). MD buckles may be throughout a roll, but more commonly occur in a particular region (e.g. near the core, on one side, between gauge bands).
How To Measure	➤ MD buckles in rolls are measured by direct observation. ➤ Note width and radial positions where MD buckles occur – Full width vs. in specific lanes cross the roll's width, throughout the roll or only in specific radial positions, such as in layers near the core or at the roll's outer diameter layers.
Problems	✓ MD buckles may become permanent deformations in the web and create functional or optical defects in final products, especially in laminated or coated products.
Mechanisms And Root Causes	1. MD hygroscopic expansion of the outer layers of roll wound at low humidity, absorbing moisture in the outer layers from a humid environment. 2. MD width growth from recovering necked width as layers change there high wound-on tension to their lower (or compressive) in-roll tension. The tension loss and width recovery is driven by compression of the inner layers of a roll by the pressure of the outer layers or from any mechanism that causes a roll's layers to lose thickness over time, including entrained air escaping, coating drying, adhesive layers flowing laterally, paper core drying, and surface asperities relaxing. This defect is most common in thin film winding, especially thin films coated with hard coatings.
Solutions To Hygroscopic Expansion TD Buckles	1. Ensure moisture level in hydrophilic products is near the long-term equilibrium. If processing, such as drying, has removed moisture, add moisture back to the product. 2. Wrap rolls in moisture barrier bags or packaging. 3. Store rolls in humidity controlled environment.
Solutions To Constrained Width Recovery TD Buckles	1. Reduce air entrainment with winding nip roller, low speed, or small diameter. 2. Reduce "valleys" in roll crossweb diameter with reduced crossweb thickness variation, reduce roll buildup ratio, thickness oscillation, or lower radial modulus.

4C3: Slip Knots

Definition	Slip knots are buckled regions of a winding roll where the shear between sticking and slipping contact buckles the web. Slip knots are usually found in the outer layers of winding optically clear films with pressure-dependent coefficient of friction between the top and bottom sides of the web. Slip knots may start as dark spots that look like the layers have a 'wetted' contact. In the extreme condition, as additional layers are wound on, small slip knots may grow like a snowball forming into large odd-shaped buckles.
How To Measure	➤ Direct observation ➤ Unwind a roll with slip knots or convolution wrinkles down to where the defect first starts. Look for surface contamination or roughness variations at defect's originating layers.
Problems	✓ Visual defect ✓ May develop bond high enough to cause blocking and tear outs at unwinding. ✓ May form permanent deformation in web.
Mechanisms And Root Causes	1. Non-uniform pressure in the topmost layer of a winding roll combined with pressure-dependent coefficient of friction cause the entering web layer to stick in some areas and slide in neighboring area, creating a local shear stress and subtle buckling. 2. Layers wound on top of subtle slip knots have a slightly larger shear stress and buckled area, leading to viscous cycle where once initiated slip knots only magnify. 3. Slip knots may originate from surface contamination where a particle as small as 4-mil (100 microns) create a high friction point as it acts as a tent pole under the top winding layer.
Solutions To Slip Knots	1. Change the slip and friction characteristics of the winding roll by modifying the web (not usually an option), including coating, adding coated or internal lubricants, changing the surface roughness, adding talc or other dust-like particles. Some one-sided products have a coating or surface modification to their non-functional backside solely for the purpose of improved winding. 2. Wind the product with an interleave layer that improves the winding frictional interfaces, instead of side A to side B of the primary web the critical frictional combinations are primary web to interleave. 3. Wind with control-gap winding to reduce the span length and freedom to buckle in response to winding roll diameter variations. 4. Wind with a higher speed to tension ratio to create more lubrication from entrained air. 5. Knurl the edges of the product to reduce the pressure of contact in the center of the rolls, allowing more entrained air to hide the thickness-induced diameter variations.

Winding Improvement Action Plan

Frequent Winding Improvement Recommendations

1. Reduce roll buildup ratio by
 - a. Winding on bigger cores
 - b. Reduce roll length
2. Decrease radial modulus (relative to tensile modulus)
 - a. Make product from paper or nonwoven substrate
 - b. Coat or laminate with compressible material
 - c. Wind with compressible interleave web
 - d. Wind at lower tension and nip load
 - e. Wind at higher speed and allow more air entrainment
 - f. Emboss or roughen the web surface
3. Manage air entrained within the winding roll
 - a. Use winding nip when increasing radius and V/T lubricate winding
 - b. Use enough winding nip to reduce air layer/thickness to less than strain
4. Optimize starting tension and tension taper (final tension)
 - a. Ensure tight roll start
 - b. Taper winding tension: 20-50% for film winding, constant torque for buildup ratio < 3
 - c. In taper experiments, vary starting tension, but keep final tension constant.
5. Align winding roll
 - a. Align winding shaft
 - b. For turret winders, align the turret pivot axis
 - c. Minimize shaft deflection, especially in cantilevered and slip shaft winding
 - d. Use cylindrical cores and engage core concentrically
6. Design product friction for wind-ability
 - a. Avoid low friction products (COF < 0.15)
 - b. Avoid high friction products (COF > 0.6)
 - c. Avoid products with pressure-dependent friction (non-Coulomb)
7. Minimize crossweb thickness variations
8. Manage entry span
 - a. Avoid free span winding where the roll controls a long upstream span
 - b. Use control gap winding for high friction, adhesive, and low speed winding
 - c. Use nipped winding for low friction and high speed winding
9. Ensure a good roll start (wrinkle-free and well-aligned)
10. Choose of core based product and process needs
 - a. Proper core strength prevents core crushing
 - b. Match core stiffness to product stiffness to minimize blocking and starring
11. Minimize roll storage time
12. Wind and store in a clean environment

V. Acknowledgments / References

David Roisum (www.roisum.com), Finishing Technologies – David has extensive experience in paper finishing processes from his work experience at Beloit (a paper winding equipment manufacturer, now out of business but with thousands of winders still in use) and Kimberly-Clark (maker of tissue, paper towels, napkins, toilet paper, diapers, and other paper and nonwoven products). Dave has been an independent consultant since 1992, has a PhD from Oklahoma State University, and resides in Neenah, WI. When in doubt, consult any of Dave's several books on web handling and winding, including the "Mechanics of..." series (Web Handling, Rollers, and Winding) and his latest book (co-authored with Dr. Keith Good of the OSU WHRC), Winding: Mechanics, Machines, and Measurements (which includes the winding model Excel spreadsheet, WindARoll).

Dilwyn Jones (www.emral.com), Emral Ltd. – Dilwyn was the web handling and winding expert for polyester films at ICI in the United Kingdom. Dilwyn is one of the driving forces behind Rheologic TopWeb web handling and winding software. He has a PhD in Physics from University of Cambridge and has been an independent consultant since 2000.

Steve Abbott, Consultant – Steve is the master programmer and author of the TopWeb web handling and winding software. Steve is the former Research Director at MacDermid Autotype, a manufacturer of coated films for electronic displays, digital printing, in-mold printing, and other specialty applications. Steve is also a visiting Professor at Leeds University in the UK, where most of the rest of the Rheologic TopCoat team are found.

Kevin Cole, Optimation – Kevin is the lead engineer of Optimation's Media Conveyance Facility. Optimation (Rochester, NY) has possibly the world's best experimental and analytical web handling facility (the Media Conveyance Facility of MCF, formerly the web handling research lab of Kodak), including limited production capabilities, equipment and development engineering support, custom machine capability, and training services.

Jerry Brown, Essex Systems – Jerry Brown is an independent consultant working on advanced understanding of web handling dynamics. Formerly of Fife Corp., Jerry works on a "wide range of both mechanical and electronics projects involving automatic control, precision mechanisms, web process dynamics, motion control, sensors, signal processing, data acquisition and computer modeling" (notes from his website).

Gerald Kheboian – Gerry is a consultant on web transport systems; has over 45 years of experience on web handling, drives, tension controls, and machine controls. Former Polaroid Technical Manager. For these notes, I consulted his section of the book Coating and Drying Defects: Troubleshooting Operating Problems (by Edgar Gutoff and Edward Cohen), Chapter 11: "Problem Associated with Web Handling," covering the importance of web handling to good coating and drying. This chapter is almost 70 pages long with over 20 figures and tables.

3M Web Handling Group Experts (past and present) – I can't forget to mention the many 3Mers that I worked with during my 17 years at 3M, including Doug Kedl, Ron Swanson, Jim Dobbs, and Dan Carlson, plus many web handling controls specialists in 3M Engineering Process Instrumentation and Controls Engineering group (or whatever they are called today), including Howie Bloom, Steve Cinnamon, Glen Copeland, and Ron Dennis.